
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

Anonymous authors
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ABSTRACT

Algorithms increasingly function as information mediators – ranging from social-media feeds and targeted advertising to the growing deployment of LLMs. This induces a coupled decision process in which each agent integrates a private, algorithmically-mediated signal with observational inference drawn from the actions of its predecessors. To formalize such settings, we introduce a model of *controlled sequential social learning* in which an information-mediating planner (e.g., an LLM) parameterizes the information structure supplied to a sequence of agents who simultaneously perform Bayesian inference over the observed decisions of earlier agents. The planner’s objective is either to maximize aggregate social welfare (the *altruistic* planner) or to induce a fixed planner-preferred action irrespective of the state (the *biased* planner). The framework yields a novel optimization problem that couples dynamic programming for the planner with decentralized, utility-maximizing action selection and Bayesian belief updates by the agents. Within this formulation, we establish convexity of the optimal value function and derive closed-form characterizations of the optimal altruistic and biased policies; these policies realize the state-dependent trade-off between the planner’s control cost and the payoff accrued from induced agent actions. The characterization shows that the optimal policy is piecewise-defined over the public-belief interval, operating in distinct regimes: investing the maximum admissible precision, investing none, or setting precision as a monotonically increasing or decreasing function of the belief. Notably, over certain belief sub-intervals the biased planner optimally attenuates signal precision, thereby obfuscating the underlying state. We further show that, even under stringent transparency constraints – informational parity with the agents, prohibition of falsification and cherry-picking, and full observability of the control – information mediation can shift aggregate welfare substantially in either direction. We complement the theory with simulations instantiating LLMs as both planner and agents. The LLM planner exhibits emergent strategic behavior in steering the public belief that closely tracks the predicted policy structure; the residual deviations are attributable to systematic non-Bayesian belief updating consistent with documented cognitive patterns of human decision-makers and of LLMs trained on human-generated data. Collectively, these results establish the framework as a tractable and empirically grounded basis for analyzing and regulating LLM information mediators.

1 INTRODUCTION

Large Language Models (LLMs) are increasingly deployed as information mediators in socially-critical functions, ranging from search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) to personal assistants that issue medical or political advice (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). Alongside this promise, LLMs have been shown to exhibit persuasive (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulative (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) behavior in their interactions with users. At population scale, this algorithmic influence does not act in isolation: it is superimposed on the endogenous diffusion of information that arises as individuals observe and infer from one another’s actions, producing coupled social dynamics that can either amplify or attenuate the model’s effect.

As a concrete instantiation, consider an LLM-powered recommendation system that presents a new restaurant to a sequence of prospective customers. The restaurant’s quality—the latent binary state of the world—is either good or bad and is unobserved by both the users and the system. Each user receives a personalized recommendation (a private signal) and then selects a binary action: whether or not to patronize the restaurant. The recommendation system (the *planner*) can expend resources to increase the informativeness of this signal—for instance, by prompting the LLM to generate a review tightly tailored to the user’s known preferences. This expenditure carries a concrete cost, corresponding to investments in market research, user surveys, or data acquisition required to tailor information effectively.

Critically, however, a user’s decision is not a function of the private signal alone. Each user additionally conditions on the observed actions of its predecessors—if the preceding ten users elected to patronize the restaurant, the next user rationally infers that they likely received favorable signals. This mechanism, termed *social learning*, is a potent determinant of collective human opinion across diverse domains, including vaccine adoption (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), technology adoption (e.g. Gillingham & Bollinger (2021); Weber (2012)), and the formation of political and moral opinions (e.g., Brady et al. (2021); Guilbeault et al. (2018)). The central contribution of this work is to formalize and analyze the strategic interaction between a planner that controls the precision of private signals and a population of agents that perform social learning. The planner must accordingly account not only for its instantaneous effect on the current user but also for the information *externality* it induces—namely, the perturbation to the public belief that conditions the decisions of all subsequent users.

While real-world interactions are highly complex, we abstract this dynamic into a tractable formal model of *controlled social learning*, in which an algorithmic planner selects, at a cost, the precision of the private signal supplied to each agent in the sequence. The planner’s mode of influence is therefore restricted: it does not falsify or fabricate information, but only determines the resources allocated to rendering each signal clear and informative. Our analysis addresses three questions that this formulation raises: by what mechanism can a planner steer the collective belief; how should the optimal control adapt to the evolving public belief; and what is the resulting effect on aggregate social welfare?

The answers depend critically on the planner’s objective. An *altruistic* planner—modeling an ideal recommendation system—seeks to maximize agents’ expected utility by inducing the state-matching action (patronize the restaurant if and only if it is good). By contrast, a *biased* planner—modeling, e.g., a system compensated by the restaurant—seeks to induce a fixed action (patronize) independent of the true state.

This distinction is related to, but formally distinct from, the notion of alignment. The altruistic objective is, by construction, aligned with maximizing each agent’s *expected utility* under state uncertainty. The biased objective, being state-independent, is not; nevertheless, its induced actions are not uniformly detrimental. When the realized state favors the planner’s preferred action (e.g., the restaurant is in fact good), the biased planner’s influence coincides with the agent’s *realized utility* in that instance. We characterize the full spectrum of these interactions, analyzing both altruistic and biased planners in regimes where the planner’s objective is aligned or misaligned with the agents’.

Contributions and Outline This paper makes the following principal contributions:

1. **A Formal Framework for Controlled Social Learning.** We introduce, to our knowledge, the first formal model that couples a dynamic control problem for a centralized information planner with the mechanism of sequential social learning. The planner selects the precision of each agent’s private signal so as to optimize an objective that is either altruistic (maximizing aggregate social welfare) or biased (inducing a fixed action).
2. **A Rigorous Characterization of Optimal Planner Policies.** We derive closed-form characterizations of the optimal altruistic and biased policies as functions of the evolving public belief. The altruistic characterization rests on a novel proof that the optimal value function is convex in the public belief. These results make explicit the strategic trade-offs the planner faces when its control induces an information externality on future agents.
3. **Empirical Validation and Strategic Analysis via LLMs.** We conduct simulations that instantiate LLMs as both planner and agents, yielding three principal findings: (a) a planner

that accounts for social learning attains substantially higher utility, and exerts substantially greater influence on the public belief and social welfare, than a myopic planner; (b) the emergent LLM-planner policy closely matches the analytically optimal policy, indicating robustness of the characterization to non-Bayesian agent behavior; and (c) LLMs display sophisticated strategic reasoning, underscoring both their capability and the attendant societal risks.

Section 2 surveys related work; Section 3 specifies the formal model; Sections 4 and 5 derive the optimal altruistic and biased policies, respectively; Section 6 presents the LLM-based empirical evaluation; and Section 7 concludes. Full proofs and supplementary results are deferred to the appendices.

2 RELATED WORK

Our work draws on four literatures: social learning, information design, online persuasion and reinforcement learning (RL), and LLMs.

Social Learning Bikhchandani et al. (1992) and Banerjee (1992) introduced the sequential social-learning framework in economics, and subsequent work has characterized the induced dynamics under progressively more general assumptions (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)). This literature has further analyzed variations in the social information available to agents and a range of agent biases and misperceptions (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010), as well as Bikhchandani et al. (2024) and references therein).

The literature on the *control* of social learning is comparatively sparse (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021). The most closely related works are Wei & Anastasopoulos (2022) and Smith et al. (2021). Wei & Anastasopoulos (2022) assumes bidirectional communication between planner and agents (agents report their private signals to the planner in exchange for action recommendations). Smith et al. (2021) considers a benevolent planner that directly modifies agents’ choice rules and derives a decentralized pivot mechanism to accelerate learning. Our model assumes neither bidirectional communication nor planner control over agent actions: agents retain full autonomy over their action selection. This makes the model better suited to information-mediating algorithms, which are typically opaque or entirely invisible to the users they serve.

Information Design In our setting, the planner optimizes its objective by selecting the precision of each agent’s private signal. This constitutes a constrained instance of the classical information-design problem (Bergemann & Morris, 2019). Such models were established by Kamenica & Gentzkow (2011)’s formulation of Bayesian persuasion, in which a sender commits to a signal structure that informs a rational Bayesian receiver about a latent state. Our planner thus solves, in each period, a cost-constrained information-design problem restricted to binary symmetric channels, where deviations from a baseline precision incur a cost. Relative to the many existing variants of information design, the distinguishing feature of our formulation is the incorporation of social learning across the sequence of agents.

Few works address information design within sequential social-learning settings (Arieli et al., 2022; Wu et al., 2025). Both consider a one-shot design in which the sender commits at the outset to a single information structure held fixed across the entire agent sequence. Arieli et al. (2022) studies a self-interested sender that selects a general information structure, as in classical Bayesian persuasion. Wu et al. (2025) restricts the information structure to be Blackwell-more-informative and maximizes the probability of a correct cascade. In contrast, we study a dynamic problem in which the sender selects a distinct signal structure for each agent, admitting a strictly richer class of sender strategies.

Online Persuasion and RL A growing body of work sits at the interface of online learning and Bayesian persuasion/information design. One line studies online sequential persuasion under unknown environment parameters, in which a sender interacts repeatedly with short-lived receivers without knowledge of the prior over states and/or the receivers’ utility functions (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). A related line examines sequential persuasion in dynamic environments or MDPs, introducing state transitions and/or receiver foresight; these models employ history-dependent or promise-based signaling

policies and frequently connect to reinforcement learning (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). Across this literature, the analytical objective is a learning-theoretic guarantee—typically sublinear regret against the best signaling scheme in hindsight and/or approximation algorithms under computational constraints.

By contrast, our model requires no parameter learning by the planner: the sequential dependence originates in social learning among receivers rather than in the sender’s updating of beliefs about the environment. Moreover, the planner holds no informational advantage—it observes neither the latent state nor the private signals. Within this regime we obtain exact characterizations of the planner’s optimal policies. Although sequential, the setting is neither an online-learning nor an RL environment: the planner’s actions do not alter the payoff-relevant state (there are no reward externalities), and the planner does not estimate unknown environment parameters. The planner refines its information solely by observing agents’ actions, which encode social learning and belief propagation rather than payoff exploration or state transition. The sequential coupling therefore arises exclusively from information externalities, whereby each agent’s action perturbs the beliefs of its successors.

LLMs A recent line of work studies LLMs specifically in the context of information design (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). These works are confined to one-to-one sender–receiver interactions and thus do not capture the social dynamics we investigate. The most closely related is Duetting et al. (2025), which studies information design where the sender jointly selects a signal structure and a ‘framing’ that conditions the (possibly non-Bayesian) posterior belief of the receiver, and employs an LLM-based oracle to solve the computationally intractable optimization over the framing space. This approach rests on the premise that LLMs can serve as behavioral proxies for human economic agents (Dillion et al., 2023; Horton, 2023). Their formulation does not address our central concerns of repeated interaction and social learning; we nonetheless adapt components of their methodology for the empirical study in Section 6.

3 SOCIAL LEARNING MODEL

We consider a planner and a countably infinite sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature draws a fixed, *unobserved* exogenous state of the world $\omega \in \Omega := \{G, B\}$ with $\mathbb{P}(\omega = G) = b_1$, where the prior b_1 is common knowledge. In the recommendation-system instance of Section 1, the state G denotes a good restaurant.

At each time i , the planner determines the level of personalization allocated to agent i and, consequently, transmits to agent i a private signal $s_i \in \Omega$ that informs its decision. The signal is binary and is generated by a binary symmetric channel: it matches the true state with probability $q_i \in [0.5, 1]$, i.e., $\mathbb{P}(s_i = \omega) = q_i$. We refer to q_i as the *signal precision* of agent i ; it quantifies the signal’s informativeness about ω .

Conditional on ω , agent i ’s signal is independent of the signals of all other agents and of the history. Conditioning on this signal and on the observed actions of the predecessors $j < i$, agent i selects an action $a_i \in \Omega$. The agent’s utility is 0 if its action matches the true state ($a_i = \omega$) and $-C$ otherwise, with $C > 0$; each agent selects the action that maximizes its own expected utility. In the running example, this corresponds to patronizing the business if and only if the state is G .

3.1 AGENTS’ DECISION PROBLEMS

Each agent observes the actions of all its predecessors together with their signal precisions. We denote this history by $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Since the planner and all agents share the identical information set $\mathcal{H}_i$, there is a common *public* belief about ω that is updated after each agent acts. The public belief immediately preceding agent i ’s action is $b_i := \mathbb{P}(\omega = G | \mathcal{H}_i)$; prior to any action it equals the prior b_1 . As in the canonical models of Banerjee (1992) and Bikhchandani et al. (1992), the sequence $(b_i)_{i \in \mathbb{N}}$ is a Markov process and constitutes a sufficient statistic for the history $\mathcal{H}_i$ (proof in Appendix B.3).

Conditioning on $\mathcal{H}_i$ (equivalently, by the Markov property, on b_i) and on q_i , agent i selects the action $a_i \in \Omega$ that maximizes its expected utility, breaking ties in favor of the action matching s_i . Agent

i forms a *private* belief $\tilde{b}_i$ about ω by combining $\mathcal{H}_i$ with its private signal s_i of precision q_i via Bayes' rule (derivation in Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

Given this private belief, the agent selects the action corresponding to the maximum-a-posteriori state under $\tilde{b}_i$: it sets $a_i = G$ if $\tilde{b}_i > 0.5$, $a_i = B$ if $\tilde{b}_i < 0.5$, and $a_i = s_i$ if $\tilde{b}_i = 0.5$. Substituting the expression for $\tilde{b}_i$ yields the following action rule:

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Agent i 's action is subsequently observed by all later agents and incorporated into the updated public belief b_{i+1} according to the following transition (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. When $1 - q_i \leq b_i \leq q_i$, Equation (2) implies that agent i 's action perfectly reveals its private signal, so the updated public belief coincides with agent i 's private belief. Otherwise, Equation (2) shows that the private signal is uninformative for the action, the action carries no information about ω , and the public belief is left unchanged; the process has entered an absorbing state, termed an *information cascade* or *herd* in the classical social-learning literature (Bikhchandani et al., 1992; Banerjee, 1992). Once such a state is reached, the public belief ceases to aggregate agents' private information.

Agent i 's expected utility equals $-C \mathbb{P}(a_i \neq \omega | b_i, q_i)$, which admits the following closed form (see Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega | b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

We consider two planner types: (1) an *altruistic* planner, which seeks to induce the state-matching action ($a_i = \omega$); and (2) a *biased* planner, which seeks to induce a fixed action—say G —independent of ω . We index these types by the subscripts A and B , respectively. In both cases the planner selects the precision of each agent's private signal, and the control cost is specified by a function $\beta(\cdot)$ that is non-negative, increasing, continuous, and concave in its argument. The planner's information set is $\mathcal{H}_i$, identical to that of the agents.

Altruistic Planner At time i , the altruistic planner selects the precision q_i for agent i and incurs cost $\beta(q_i)$, where $\beta(p) = 0$ for the baseline precision $p \in [0.5, 1)$. For the altruistic planner, reducing precision below baseline is never beneficial (Appendix C.9); accordingly, and to streamline notation, the planner incurs a positive cost only when it raises precision above p , with the cost increasing in precision at a decreasing marginal rate (reflecting concavity of β). Both p and $\beta(\cdot)$ are common knowledge among the agents.

The altruistic planner maximizes social welfare—defined as the expected total utility of the agents—net of the cumulative cost of precision investment. Let $r_A(b_i, q_i)$ denote the instantaneous reward accrued by the altruistic planner at public belief b_i under precision q_i for agent i . Starting from prior b_1 and following a policy sequence $\pi = (\pi_i)_{i=1}^\infty$ with $\pi_i(\mathcal{H}_i) = q_i$, the planner attains the following expected total discounted return under discount factor $\delta \in [0, 1)$:

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The optimal value and optimal policy of the altruistic planner are the supremum and arg-supremum of $V_A^\pi(\cdot)$ over $\pi \in \Pi$, where Π denotes the set of admissible policies. The myopic value and policy are recovered as the special case $\delta = 0$.

Biased Planner The biased planner’s problem differs from the altruistic planner’s only in the objective, and hence in the cost and reward functions; Appendix A develops a political-campaign instance that illustrates the biased planner. The biased planner seeks to induce action G from every agent, independent of ω : it incurs cost 0 when an agent selects G and cost $C > 0$ when an agent selects B , in both cases irrespective of ω . Unlike the altruistic planner, the biased planner may set precision either above or below the baseline p , and both directions of deviation are costly. Any deviation from p requires tailoring the signal to the specific agent—which in turn requires research into how that agent best assimilates content—and therefore incurs a cost. Concretely, the biased planner incurs cost $\beta(|q_i - p|)$ for precision q_i , with $\beta(0) = 0$.

The biased planner’s expected instantaneous reward at time i is accordingly defined as:

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C\mathbb{P}(a_i = B|b_i, q_i) \quad (6)$$

Substituting Equation (6) into the return, the biased-planner objects $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and $\pi_B^*(\cdot)$ are defined exactly as $V_A^\pi(\cdot)$, $V_A^*(\cdot)$, and $\pi_A^*(\cdot)$ were defined for the altruistic planner from Equation (5).

As stated in the opening paragraph of Section 3, each agent obtains strictly higher utility from the state-matching action. Consequently, when $\omega = B$, the biased planner’s success in inducing G strictly reduces the agent’s utility. The altruistic planner, by contrast, induces the state-matching action for every realization of ω and therefore always increases the agent’s utility. Hence the altruistic objective is aligned with every agent’s objective unconditionally, whereas the biased objective is aligned only on the event $\omega = G$.

Both planners’ problems are infinite-horizon discounted stationary Markov Decision Processes (MDPs) with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition kernel given by Equation (3) (Puterman, 1990). Each therefore admits a unique optimal value function (Kallenberg, 2011). We restrict attention to deterministic Markov policies, i.e., $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ with $\pi(b_i) = q_i$.

Remark 2. *Our model rests on three substantive assumptions: (1) Informational parity. The planner’s information set coincides with the agents’. This is restrictive when the planner commands richer data, but non-restrictive when all parties condition on the same public signals, e.g., early findings or survey data. (2) Binary symmetric channel. The planner modulates the signal precision, but every signal is transmitted through a binary symmetric channel. This is restrictive if the planner can cherry-pick, censor, or falsify content, but non-restrictive when its role is limited to improving the quality of noisy sources without altering their content. (3) Full observability of the control. The planner’s precision choices are fully observable. This is restrictive when covert policies or framing are feasible, but non-restrictive in transparent environments.*

4 OPTIMAL ALTRUISTIC POLICIES

We first analyze the myopic case, which disregards the effect of social learning on future agents. In our formulation this corresponds to the discount factor $\delta = 0$, under which the planner discounts all future rewards to zero and $V_A^\pi(b) = r_A(b, \pi(b))$. The optimal myopic policy $\pi_A^0(\cdot)$ is then defined pointwise as

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

The myopic altruistic problem admits an equivalent decentralized formulation: each agent selects the precision of its own private signal, incurs the associated cost, and maximizes the sum of its expected utility and negated cost, independently of other agents.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic optimal policy is therefore of threshold form. When the public belief is sufficiently far from 0.5, the planner selects the baseline precision p at zero cost; when the belief is near 0.5, the planner supplies a perfect signal (precision 1). The threshold t_M depends only on the cost of a perfect signal, $\beta(1)$, and the cost of an incorrect action, C . When $\beta(1) \geq C$, the perfect signal is

prohibitively expensive relative to the mistake cost and is never deployed; consequently $t_M = 0.5$ and the belief interval on which the myopic policy selects precision 1 is empty.

We now establish a structural property of the altruistic optimal value function that is central to the subsequent analysis:

Theorem 2. $V_A^*(\cdot)$ is convex with respect to public belief.

The proof of Theorem 2 (Appendix C.3) is technically involved and may be of independent interest. The principal difficulty is the dependence of the agents' actions on the public belief. Were the actions independent of the public-belief process (as in Nyarko (1994)), the expected utility would be a linear function of the belief state and convexity of the value function would follow immediately; here this argument does not apply.

Convexity of the value function is the key structural property enabling the characterization of the optimal policy.

Theorem 3. There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A) \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & \text{o.w.} \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal policy partitions the public-belief interval into three regimes. First, as in the myopic case, the optimal policy withholds precision investment at extreme beliefs. Notably, the non-myopic optimum requires a more extreme belief than the myopic optimum before it ceases investing, precisely because it internalizes the effect on future agents whereas the myopic policy does not.

Second, when the public belief lies near 0.5 and is thus minimally informative, the optimum selects precision 1 whenever this is not cost-prohibitive. By Equation (2), the agent's action then matches the true state with probability 1, and the public belief collapses to 0 or 1.

Third, on the remaining belief interval the optimum selects the minimal precision $\max(b, 1 - b)$ for which the agent's action reflects its private signal (see Equation (2)). Any lower precision renders the action uninformative relative to the incumbent public belief; the chosen precision is thus the minimum-cost level that sustains social learning through peer observation.

5 OPTIMAL BIASED POLICIES

As in Section 4, we begin with the myopic optimum. Paralleling that treatment, the myopic biased optimal policy $\pi_B^0(\cdot)$ is the arg-supremum of the instantaneous reward $r_B(b, q)$ over q . We now characterize $\pi_B^0(\cdot)$, noting that—unlike the altruistic case—an optimal policy need not exist in the biased case, as the supremum may fail to be attained (Puterman, 1990, sec. 2.3.1). Where the optimum is unattained, we instead characterize ϵ -optimal policies, denoted by an ϵ superscript.

Theorem 4. There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:

(A) If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.

(B) If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.

(C) If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.

(D) If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.

Proof in Appendix C.6. We now leverage Theorem 4 to characterize the (non-myopic) optimal biased policy.

Theorem 5. There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:

(A) If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.

(B) If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.

- 378 (C) If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$
 379
 380 (D) If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.
 381
 382 (E) If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.

383 Theorem 5 (proven in Appendix C.8 and illustrated in figure 2a) partitions the belief interval into
 384 five regimes, which we interpret in turn.

385 *Regime (A): extreme beliefs.* When the public belief is sufficiently far from 0.5, the cost of steering
 386 the system exceeds its benefit—even though a low belief ($b < 0.5$) is unfavorable to the planner. The
 387 selected precision p satisfies $p < \max(b, 1 - b)$, so by Equation (2) each agent acts on the public
 388 belief irrespective of its private signal. The action is uninformative, the public belief is invariant,
 389 and—since the policy is Markovian—this repeats indefinitely: an unfavorable cascade if $b < 0.5$
 390 and a favorable one otherwise.

391 *Regime (B): near an unfavorable cascade.* The planner raises precision to a level sufficient to make
 392 the agent’s action responsive to its private signal, even though the signal realizes as B in expectation.
 393 The agent then acts on its private signal, which realizes as G with strictly positive probability—a
 394 last-resort investment aimed at escaping the unfavorable cascade.

395 *Regime (C): belief slightly higher.* The planner may lower precision below p . Here $b < 0.5$, so
 396 higher precision more likely yields unfavorable (B) news; the planner therefore retains precision
 397 just large enough to keep the action responsive to the signal ($q \geq 1 - b$), seeking a more favorable
 398 belief while using the least informative admissible signal.

400 *Regime (D): belief weakly favorable.* The planner adopts precision at least p . Since $b > 0.5$, raising
 401 precision increases the probability that the agent infers $\omega = G$; investment in this regime serves to
 402 reinforce the favorable public belief.

403 *Regime (E): belief strongly favorable.* The planner lowers precision just below $\max(b, 1 - b)$, to
 404 $b - \epsilon$, so that agents disregard their private signals and select G . Here the risk that an informative
 405 private signal overturns the favorable belief outweighs both the cost of attenuating precision and the
 406 prospective gain from further increasing the belief.

408 6 EVALUATION VIA LLM-BASED SIMULATION

410 We now employ LLMs to empirically assess the implications and external validity of the model and
 411 its theoretical predictions. The evaluation proceeds in three stages: we (i) characterize LLM-agent
 412 behavior to quantify its deviations from Bayesian rationality, (ii) measure how an LLM planner
 413 adapts its policy in response, and (iii) quantify the resulting effect on social welfare.

414 We operationalize the controlled social-learning setting in a scenario in which agents decide whether
 415 to purchase a new car. As depicted in figure 1a, LLMs instantiate three roles, each specified below
 416 with prompts given in Appendix E.2.2:

417 **Agent:** conditioned on an assigned profile, the agent observes the history of prior actions and a
 418 private message about the car of known precision, forms a posterior belief over the car’s quality, and
 419 outputs a binary buy/not-buy decision.

420 **Planner:** conditioned on the action history, the planner selects the precision q_i of agent i ’s private
 421 signal in accordance with its objective.

422 **Oracle:** given the planner’s chosen precision, an agent profile, and a car fact sheet (Appendix E.2.1),
 423 the oracle generates a profile-tailored private signal realizing the specified precision. We benchmark
 424 the LLM-based simulations against numerical solutions of the planner’s MDP. In both the LLM and
 425 numerical experiments we adopt a linear cost function $\beta(q) = k|q - p|$ and sweep the cost scale
 426 k , the baseline precision p , and the discount factor δ , holding the mistake cost fixed at $C = 1$.
 427 Appendix E details the experimental setup and prompts; Appendix E.3 validates both the elicited
 428 beliefs and the oracle’s precision-generation performance.

429 The experiments support three conclusions:

- 430 1. **Impact of social-learning control.** Both the analytical and the LLM planners attain sub-
 431 stantially higher utility by internalizing and exploiting the social-learning dynamics. The

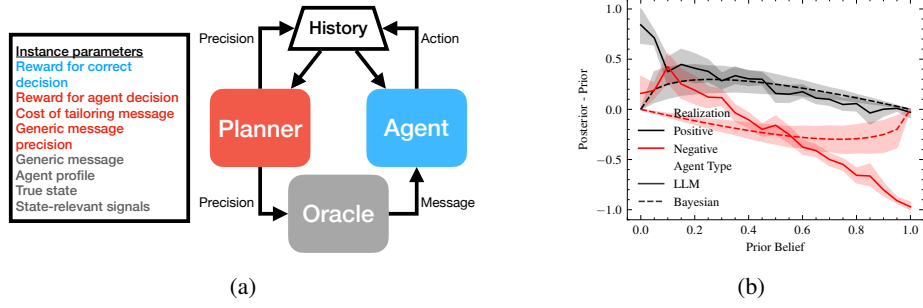


Figure 1: (a) System diagram of the experimental pipeline. Instance parameters are color-coded by the LLM role that consumes them. (b) Posterior belief of LLM agents (solid) and Bayesian agents (dashed) as a function of a fixed prior, following a positive (black) or negative (red) private signal.

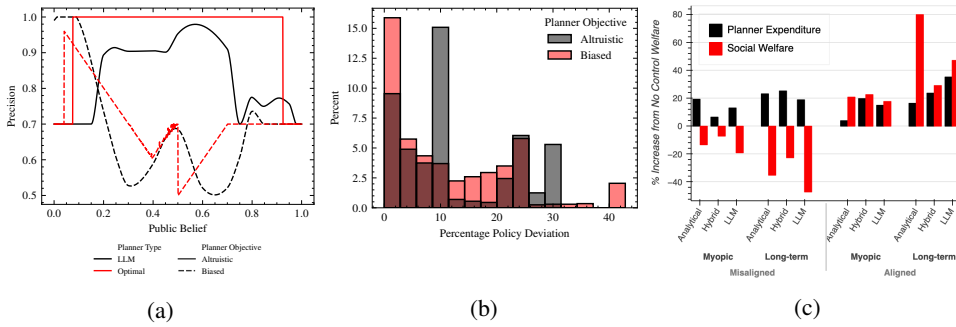


Figure 2: (a) Representative policies of the LLM planner (black) and the analytically optimal planner (red), under the altruistic (solid) and biased (dashed) objectives, as functions of the public belief. (b) Empirical distribution of the percentage deviation between the LLM and optimal policies. (c) Planner expenditure and change in social welfare, each expressed as a percentage of the no-control baseline welfare, with the true state fixed to B . The left half corresponds to a biased planner targeting action G ; the right half, to an altruistic planner.

corresponding effect on social welfare is positive or negative according to whether the planner’s and agents’ objectives are aligned.

2. **Robustness of the analytical characterization.** The LLM planner selects policies that closely track the (non-trivial) analytically optimal policies, despite interacting with non-Bayesian LLM agents.
3. **Emergent strategic behavior of the LLM planner.** The LLM planner exhibits sophisticated emergent strategy: its policies not only approximate the optimum but also deviate from it in a direction that compensates for the non-Bayesian belief updating of the LLM agents it faces.

6.1 LLMs AS NON-BAYESIAN AGENTS

Although our analysis assumes Bayesian agents, both humans and LLMs are documented to exhibit systematic departures from Bayesian updating (Rupprecht et al., 2025). To isolate these effects in the LLM agent, we measure its belief update as a function of the incoming signal. The results, reported in figure 1b, exhibit the following systematic non-Bayesian patterns in the LLM’s belief update (see also Appendix E.4):

- (NB1) The LLM agent underreacts to private signals that are concordant with its prior belief.
- (NB2) The LLM agent overreacts to private signals that are discordant with its prior belief.
- (NB3) Consequently, the LLM agent requires a more extreme public belief than a Bayesian agent to enter an information cascade.

These biases have likewise been documented empirically in human subjects (e.g., Ba et al. (2022); Chan et al. (2025)). The subsequent sections establish that the LLM planner’s policy is robust to non-Bayesian agents of this kind.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We now compare the LLM planner’s policies against the optimal policies derived in Sections 4 and 5. As shown in figure 2a, and notwithstanding the agents’ departures from Bayesian behavior, the emergent LLM-planner policy is structurally close to the theoretical optimum. This holds even when the optimum is highly non-trivial (e.g., the biased policy in figure 2a), indicating that the LLM planner engages in sophisticated strategic behavior that both accounts for and exploits social learning.

Under the altruistic objective, both planners halt investment at extreme beliefs and invest maximally near 0.5. Under the biased objective, both reproduce the same qualitative structure: heavy investment to escape an unfavorable belief, attenuated precision near the midpoint, and no investment once a favorable belief is secured. The magnitude of the policy deviation is typically small (figure 2b): for both objectives it is below 10% over the majority of belief states, corroborating the close structural agreement between the emergent LLM policy and the theoretical characterization.

There remain, however, systematic structural differences, which are most parsimoniously explained as the planner’s adaptations to the specific non-Bayesian behaviors identified in Section 6.1. (1) The LLM planner avoids the extreme precisions 0.5 and 1.0, consistent with a documented central-tendency bias (Rupprecht et al., 2025). (2) The LLM planner tapers its investment gradually rather than at a sharp threshold. This directly counteracts the agents’ cascade resistance (NB3): because even agents with strong priors can be reversed by a countervailing private signal, no belief is entirely “safe” at which to cease investing. (3) Analogously, the biased LLM planner continues to invest at very low beliefs, reflecting the agents’ propensity to overreact (NB2) to a surprisingly positive signal, which renders a last-resort intervention more viable than under Bayesian agents.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

We define social welfare $W^\pi(b)$ as the total expected discounted utility of all agents. Because additional information never decreases an agent’s expected utility, welfare is monotonically non-decreasing in signal precision (Appendix C.9); it follows that the altruistic planner weakly increases welfare relative to the no-control baseline. To quantify these effects, we compare social welfare and planner expenditure across three configurations: (1) the *analytical* configuration (optimal policy, Bayesian agents); (2) the *LLM* configuration (LLM planner, LLM agents); and (3) a *hybrid* configuration (optimal policy, LLM agents).

The results in figure 2c confirm that the planner materially alters social welfare across all three configurations, and that neglecting social learning (the myopic case) substantially degrades the planner’s outcome. In particular, when misaligned, the biased analytical and LLM planners reduce social welfare by 40 to 50%, achieving this by deliberately obscuring information about the true state via precision-attenuating policies. The magnitude of this welfare loss is especially notable given the stringent constraints imposed on the planner (Remark 2), and it substantiates the risk posed by misaligned LLM information mediators.

The results further quantify the cost of a misspecified agent model. In the hybrid configuration, the analytically optimal policy—derived under the Bayes-rationality assumption—is brittle: its performance degrades when applied to non-Bayesian agents. The LLM policy, by contrast, closely resembles the analytical policy yet is better calibrated to non-Bayesian agents exhibiting human-like biases.

7 CONCLUSION

We introduced a formal framework for the strategic control of social learning by an algorithmic information mediator and developed a tractable model of this increasingly prevalent dynamic. We established that even a tightly constrained planner can substantially shift social welfare in either direction, and that internalizing social learning is essential to exercising this influence effectively.

Our simulations corroborated these findings: LLMs perform strategic reasoning whose structure matches our analytical predictions while adapting to non-Bayesian agents.

Collectively, these findings yield three takeaways for algorithmic information mediation. (1) The effect of human–AI (or human–algorithm) interaction cannot be assessed in isolation: neglecting its coupling with concurrent processes such as social learning misrepresents its potential, whether beneficial or harmful. (2) Information mediators, even when constrained to subtle influence, wield substantial power to guide or derail social learning and public opinion, motivating the study of mechanisms to mitigate the associated risks. (3) Modern LLMs exhibit emergent strategic behavior that both accounts for and exploits social learning and non-Bayesian cognitive biases. The latter two points underscore the need to better understand and mitigate the risks of algorithmic information mediators, and of LLMs in particular. A principal limitation of our study is the absence of human-subject data; given the ongoing debate over the fidelity of LLM-based human simulators (see, e.g., Horton (2023); Gao et al. (2025)), such data would be valuable for validating—and potentially fine-tuning—the LLM simulators.

We close by identifying two directions for future work: (1) generalizing the model and its results, and (2) mitigating the adverse welfare effects of biased planners. Several generalizations would relax our modeling assumptions. Larger state spaces, more general signal structures, more general loss functions, and heterogeneous agent populations are all natural extensions toward greater realism. Relaxing the binary-state and 0–1-loss assumptions poses primarily algebraic—rather than conceptual—difficulty, and we conjecture that the qualitative structure of our results persists. For more general signal structures, the central obstacle is preserving the convexity result of Theorem 2; identifying the broadest class of signal structures under which convexity holds is an especially interesting open problem. As a first step toward heterogeneous agents, Appendix D extends the altruistic-planner results to a population containing agents with opposing preferences.

A further important direction is the design of interventions that prevent the welfare loss induced by biased planners. This includes regulatory or mechanism-design approaches that align planner and agent incentives so as to preclude detrimental welfare effects.

REFERENCES

- Shipra Agrawal, Yiding Feng, and Wei Tang. Dynamic Pricing and Learning with Bayesian Persuasion. *Advances in Neural Information Processing Systems*, 36:59273–59285, December 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/b9c2e8a0bbcd5fcfaf62856a3a719ada-Abstract-Conference.html.
- Lisa P. Argyle, Christopher A. Bail, Ethan C. Busby, Joshua R. Gubler, Thomas Howe, Christopher Rytting, Taylor Sorensen, and David Wingate. Leveraging AI for democratic discourse: Chat interventions can improve online political conversations at scale. *Proceedings of the National Academy of Sciences*, 120(41):e2311627120, October 2023. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.2311627120. URL <https://pnas.org/doi/10.1073/pnas.2311627120>.
- Itai Arieli and Manuel Mueller-Frank. A general analysis of sequential social learning. *Mathematics of Operations Research*, 46:1235–1249, 11 2021.
- Itai Arieli, Ronen Gradwohl, and Rann Smorodinsky. Herd Design. In *Proceedings of the 23rd ACM Conference on Economics and Computation*, pp. 871–872, Boulder CO USA, July 2022. ACM. ISBN 978-1-4503-9150-4. doi: 10.1145/3490486.3538292. URL <https://dl.acm.org/doi/10.1145/3490486.3538292>.
- Cuimin Ba, J. Aislinn Bohren, and Alex Imas. Over- and Underreaction to Information. *SSRN Electronic Journal*, 2022. ISSN 1556-5068. doi: 10.2139/ssrn.4274617. URL <https://www.ssrn.com/abstract=4274617>. Publisher: Elsevier BV.
- Francesco Bacchiocchi, Matteo Bollini, Matteo Castiglioni, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion Without a Clue. *Advances in Neural Information Processing Systems*, 37:76404–76449, December 2024.
- Fengshuo Bai, Mingzhi Wang, Zhaowei Zhang, Boyuan Chen, Yinda Xu, Ying Wen, and Yaodong Yang. Efficient Model-agnostic Alignment via Bayesian Persuasion, May 2024.
- Abhijit V Banerjee. A simple model of herd behavior. *The Quarterly Journal of Economics*, 107(3):797–817, 1992.

594 Larry M. Bartels. *Presidential primaries and the dynamics of public choice*. Princeton, N.J. : Princeton
595 University Press, 1988.

596 Chris T Bauch and Samit Bhattacharyya. Evolutionary game theory and social learning can determine how
597 vaccine scares unfold. *PLoS computational biology*, 8(4), 2012.

598 Dirk Bergemann and Stephen Morris. Information Design: A Unified Perspective. *Journal of Economic*
599 *Literature*, 57(1):44–95, March 2019. ISSN 0022-0515. doi: 10.1257/jel.20181489. URL [https://](https://www.aeaweb.org/articles?id=10.1257/jel.20181489)
600 www.aeaweb.org/articles?id=10.1257/jel.20181489.

601 Martino Bernasconi, Matteo Castiglioni, Alberto Marchesi, and Mirco Mutti. Persuading Farsighted Receivers
602 in MDPs: The Power of Honesty. *Advances in Neural Information Processing Systems*, 36:14987–15014,
603 December 2023.

604 Sujoy Bhatt and Vikram Krishnamurthy. Controlled Sequential Information Fusion With Social Sensors. *IEEE*
605 *Transactions on Automatic Control*, 66(12):5893–5908, December 2021.

606 Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change
607 as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.

608 Sushil Bikhchandani, David Hirshleifer, Omer Tamuz, and Ivo Welch. Information Cascades and Social Learn-
609 ing. *Journal of Economic Literature*, 62(3):1040–1093, September 2024. ISSN 0022-0515, 2328-8175. doi:
610 10.1257/jel.20241472. URL <https://pubs.aeaweb.org/doi/10.1257/jel.20241472>.

611 Ilai Bistriz, Nasimeh Heydaribeni, and Achilleas Anastasopoulos. Informational cascades with nonmyopic
612 agents. *IEEE Transactions on Automatic Control*, 67, 9 2022.

613 William J Brady, Killian McLoughlin, Tuan N Doan, and Molly J Crockett. How social learning amplifies
614 moral outrage expression in online social networks. *Science Advances*, 7(33), 2021.

615 Carlos Carrasco-Farre. Large Language Models are as persuasive as humans, but how? About the cognitive
616 effort and moral-emotional language of LLM arguments, April 2024. URL [http://arxiv.org/abs/](http://arxiv.org/abs/2404.09329)
617 2404.09329. arXiv:2404.09329 [cs].

618 Matteo Castiglioni, Andrea Celli, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persua-
619 sion. In *Advances in Neural Information Processing Systems*, volume 33, pp. 16188–16198. Cur-
620 ran Associates, Inc., 2020. URL [https://proceedings.neurips.cc/paper/2020/hash/](https://proceedings.neurips.cc/paper/2020/hash/ba5451d3c91a0f982f103cde249bc78-Abstract.html)
621 [ba5451d3c91a0f982f103cde249bc78-Abstract.html](https://proceedings.neurips.cc/paper/2020/hash/ba5451d3c91a0f982f103cde249bc78-Abstract.html).

622 Matteo Castiglioni, Alberto Marchesi, Andrea Celli, and Nicola Gatti. Multi-Receiver Online Bayesian Per-
623 suasion. In *Proceedings of the 38th International Conference on Machine Learning*, pp. 1314–1323. PMLR,
624 July 2021.

625 Kenneth Chan, Gary Charness, Chetan Dave, and J. Lucas Reddinger. On Prior Confidence and Belief Updat-
626 ing, May 2025. URL <http://arxiv.org/abs/2412.10662>. arXiv:2412.10662 [econ].

627 Zirui Cheng and Jiaxuan You. Towards Strategic Persuasion with Language Models, September 2025.

628 Krishna Dasaratha and Kevin He. Learning from viral content. *arXiv preprint arXiv:2210.01267*, 2022.

629 Danica Dillion, Niket Tandon, Yuling Gu, and Kurt Gray. Can AI language models replace human participants?
630 *Trends in Cognitive Sciences*, 27(7):597–600, July 2023. ISSN 1364-6613, 1879-307X. doi: 10.1016/j.
631 [tics.2023.04.008](https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613(23)00098-0). URL [https://www.cell.com/trends/cognitive-sciences/abstract/](https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613(23)00098-0)
632 [S1364-6613\(23\)00098-0](https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613(23)00098-0). Publisher: Elsevier.

633 Paul Duetting, Safwan Hossain, Tao Lin, Renato Paes Leme, Sai Srivatsa Ravindranath, Haifeng Xu, and Song
634 Zuo. Information Design With Large Language Models, September 2025. URL [http://arxiv.org/](http://arxiv.org/abs/2509.25565)
635 [abs/2509.25565](http://arxiv.org/abs/2509.25565). arXiv:2509.25565 [cs].

636 Maxim Egorov, Zachary N. Sunberg, Edward Balaban, Tim A. Wheeler, Jayesh K. Gupta, and Mykel J.
637 Kochenderfer. POMDPs.jl: A framework for sequential decision making under uncertainty. *Journal of*
638 *Machine Learning Research*, 18(26):1–5, 2017.

639 Erik Eyster and Matthew Rabin. Naïve herding in rich-information settings. *American Economic Journal:*
640 *Microeconomics*, 2:221–243, 11 2010.

641 Shen Gao, Jiabao Fang, Quan Tu, Zhitao Yao, Zhumin Chen, Pengjie Ren, and Zhaochun Ren. Generative
642 News Recommendation. In *Proceedings of the ACM Web Conference 2024*, WWW ’24, pp. 3444–3453,
643 New York, NY, USA, May 2024. Association for Computing Machinery. ISBN 979-8-4007-0171-9. doi:
644 10.1145/3589334.3645448. URL <https://dl.acm.org/doi/10.1145/3589334.3645448>.

- Yuan Gao, Dokyun Lee, Gordon Burtch, and Sina Fazelpour. Take Caution in Using LLMs as Human Surrogates: Scylla Ex Machina, January 2025. URL <http://arxiv.org/abs/2410.19599>. arXiv:2410.19599 [econ].
- Kenneth T Gillingham and Bryan Bollinger. Social learning and solar photovoltaic adoption. *Management Science*, 67(11):7091–7112, 2021.
- Douglas Guilbeault, Joshua Becker, and Damon Centola. Social learning and partisan bias in the interpretation of climate trends. *Proceedings of the National Academy of Sciences*, 115(39):9714–9719, 2018.
- Keegan Harris, Nicole Immorlica, Brendan Lucier, and Aleksandrs Slivkins. Algorithmic Persuasion Through Simulation, February 2025. URL <http://arxiv.org/abs/2311.18138>. arXiv:2311.18138 [cs].
- Claudia E. Haupt and Mason Marks. AI-Generated Medical Advice—GPT and Beyond. *JAMA*, 329(16):1349–1350, April 2023. ISSN 0098-7484. doi: 10.1001/jama.2023.5321. URL <https://doi.org/10.1001/jama.2023.5321>.
- K. Hinderer. Lipschitz Continuity of Value Functions in Markovian Decision Processes. *Mathematical Methods of Operations Research*, 62(1):3–22, September 2005.
- John J. Horton. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?, April 2023. URL <https://www.nber.org/papers/w31122>.
- Bright Huo, Amy Boyle, Nana Marfo, Wimonchat Tangamornsuksan, Jeremy P. Steen, Tyler McKechnie, Yung Lee, Julio Mayol, Stavros A. Antoniou, Arun James Thirunavukarasu, Stephanie Sanger, Karim Ramji, and Gordon Guyatt. Large Language Models for Chatbot Health Advice Studies: A Systematic Review. *JAMA Network Open*, 8(2):e2457879, February 2025. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2024.57879. URL <https://doi.org/10.1001/jamanetworkopen.2024.57879>.
- Cameron R. Jones and Benjamin K. Bergen. Lies, Damned Lies, and Distributional Language Statistics: Persuasion and Deception with Large Language Models, December 2024. URL <http://arxiv.org/abs/2412.17128>. arXiv:2412.17128 [cs].
- Lodewijk Kallenberg. Markov decision processes. *Lecture Notes. University of Leiden*, 428, 2011.
- Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.
- Vikram Krishnamurthy. Quickest Detection POMDPs With Social Learning: Interaction of Local and Global Decision Makers. *IEEE Transactions on Information Theory*, 58(8):5563–5587, August 2012.
- LG AI Research. Exaone deep: Reasoning enhanced language models. *arXiv preprint arXiv:2503.12524*, 2025.
- Minqian Liu, Zhiyang Xu, Xinyi Zhang, Heajun An, Sarvech Qadir, Qi Zhang, Pamela J. Wisniewski, Jin-Hee Cho, Sang Won Lee, Ruoxi Jia, and Lifu Huang. LLM Can be a Dangerous Persuader: Empirical Study of Persuasion Safety in Large Language Models, April 2025. URL <http://arxiv.org/abs/2504.10430>. arXiv:2504.10430 [cs].
- Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports*, 14(1), August 2024. ISSN 2045-2322. doi: 10.1038/s41598-024-69032-z. URL <https://www.nature.com/articles/s41598-024-69032-z>. Publisher: Springer Science and Business Media LLC.
- Yaw Nyarko. On the convexity of the value function in Bayesian optimal control problems. *Economic Theory*, 4(2):303–309, March 1994.
- Kenneth Payne and Baptiste Alloui-Cros. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory, July 2025. URL <http://arxiv.org/abs/2507.02618>. arXiv:2507.02618 [cs].
- Yujin Potter, Shiyang Lai, Junsol Kim, James Evans, and Dawn Song. Hidden Persuaders: LLMs’ Political Leaning and Their Influence on Voters, November 2024.
- Martin L. Puterman. Chapter 8 markov decision processes. In *Stochastic Models*, volume 2 of *Handbooks in Operations Research and Management Science*. Elsevier, 1990.
- Maya Raifer, Guy Rotman, Reut Apel, Moshe Tennenholtz, and Roi Reichart. Designing an Automatic Agent for Repeated Language-based Persuasion Games. *Transactions of the Association for Computational Linguistics*, 10:307–324, March 2022.

- Neel Rao, Markus M Mobius, and Tanya Rosenblat. Social networks and vaccination decisions. *FRB of Boston*, 2007.
- Alexander Rogiers, Sander Noels, Maarten Buyl, and Tijl De Bie. Persuasion with Large Language Models: a Survey, November 2024. URL <http://arxiv.org/abs/2411.06837>. arXiv:2411.06837 [cs].
- Jens Rupperecht, Georg Ahnert, and Markus Strohmaier. Prompt Perturbations Reveal Human-Like Biases in LLM Survey Responses, July 2025. URL <http://arxiv.org/abs/2507.07188>. arXiv:2507.07188 [cs].
- Dr.-Ing. Martin Schiele, Yannick Gittmann, Stefan Ilchmann, Ante Gojsalić, Dominik Jurinčić, and Phylis Klempt. Voting Advice Applications: Implementation of RAG-supported LLMs, July 2024. URL <https://www.techrxiv.org/users/802685/articles/1187651-voting-advice-applications-implementation-of-rag-supported-llms?commit=c540a54749906f8ac761093f5dff0f514291d4b6>.
- James E. Smith and Kevin F. McCardle. Structural Properties of Stochastic Dynamic Programs. *Operations Research*, 50(5):796–809, October 2002.
- Lones Smith and Peter Sørensen. Pathological outcomes of observational learning. *Econometrica*, 68:371–398, 2000.
- Lones Smith, Peter Norman Sørensen, and Jianrong Tian. Informational Herding, Optimal Experimentation, and Contrarianism. *The Review of Economic Studies*, 88(5):2527–2554, September 2021. ISSN 0034-6527, 1467-937X. doi: 10.1093/restud/rdab001. URL <https://academic.oup.com/restud/article/88/5/2527/6149316>.
- Shih-Tang Su, Vijay G. Subramanian, and Grant Schoenebeck. Bayesian Persuasion in Sequential Trials. In Michal Feldman, Hu Fu, and Inbal Talgam-Cohen (eds.), *Web and Internet Economics*, pp. 22–40, Cham, 2022. Springer International Publishing.
- Jeremy G Weber. Social learning and technology adoption: The case of coffee pruning in peru. *Agricultural Economics*, 43:73–84, 2012.
- Xupeng Wei and Achilleas Anastasopoulos. Social learning with a self-interested coordinator. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pp. 3137–3144. IEEE, 2022.
- Marcus Williams, Micah Carroll, Adhyyan Narang, Constantin Weissner, Brendan Murphy, and Anca Dragan. On Targeted Manipulation and Deception when Optimizing LLMs for User Feedback, February 2025. URL <http://arxiv.org/abs/2411.02306>. arXiv:2411.02306 [cs].
- Jibang Wu, Zixuan Zhang, Zhe Feng, Zhaoran Wang, Zhuoran Yang, Michael I. Jordan, and Haifeng Xu. Sequential Information Design: Markov Persuasion Process and Its Efficient Reinforcement Learning, February 2022.
- Shuo Wu, Pawan Poojary, and Randall Berry. Observational Learning with a Budget, April 2025. URL <http://arxiv.org/abs/2504.19396>. arXiv:2504.19396 [cs].
- Tianyi Zhang, Faisal Ladhak, Esin Durmus, Percy Liang, Kathleen McKeown, and Tatsunori B. Hashimoto. Benchmarking Large Language Models for News Summarization. *Transactions of the Association for Computational Linguistics*, 12:39–57, January 2024a. ISSN 2307-387X. doi: 10.1162/tacl_a_00632. URL https://doi.org/10.1162/tacl_a_00632.
- Yadong Zhang, Shaoguang Mao, Tao Ge, Xun Wang, Adrian de Wynter, Yan Xia, Wenshan Wu, Ting Song, Man Lan, and Furu Wei. LLM as a Mastermind: A Survey of Strategic Reasoning with Large Language Models, April 2024b. URL <http://arxiv.org/abs/2404.01230>. arXiv:2404.01230 [cs].

A APPLICATIONS ELABORATED

Consider a sequence of individuals (agents) deciding whether to patronize a business or service provider (e.g., a restaurant, contractor, or realtor), together with a recommendation system (the planner) that serves the agents information about the business. The latent binary state is whether the business is good, which is unobserved. Both the recommendation system and the agents seek actions consistent with this state: an agent wishes to patronize the business if and only if it is good, and the recommendation system wishes to induce exactly that action. In this sense the recommendation system is an *altruistic planner*.

The recommendation system may present an agent with a highly targeted (*precise*) advertisement that conveys the business’s strengths and weaknesses in contexts relevant to the agent’s background, or with a generic or confusing advertisement that provides little decision-relevant information. A precise signal is more informative, comprehensible, and relatable, and thus more likely to drive the agent to the correct action—patronizing the business if and only if it is good. Social welfare increases with the number of agents that act correctly. However, increasing precision is costly, as it requires tailoring to the agent’s specific background. The recommendation system therefore selects the precisions so as to maximize social welfare net of these costs.

We now give a concrete instance of a *biased planner*. Consider a population of voters (agents) motivated to support the candidate most likely to prevail in their community. This “back-a-winner” motivation is known to shape electoral outcomes—for instance, it accounts for the disproportionate influence of US states holding early primaries (see Bartels (1988)). The planner is a particular candidate’s campaign and therefore seeks to induce agents to back that candidate. The binary latent state indicates whether the candidate is winning; an agent who knew this state would back the candidate if and only if he were winning.

Each agent understands its community well and could infer whether the candidate is winning (the latent state) given knowledge of his stances and policies. Accurate inference requires that this information be conveyed in the modality the agent best assimilates—e.g., audio-visual content, long-form articles supported by facts and citations, brief and focused messages, or the agent’s native language.

The campaign transmits digital advertisements (private signals) of varying precision, where precision quantifies the degree of tailoring to the agent’s preferred modality. A precise signal accurately conveys the latent state by clarifying the candidate’s policies, record, and character, enabling correct inference about whether he will win. Precision does not, however, imply support for the candidate: if the candidate is in fact losing in the agent’s community, a highly precise signal induces the agent to oppose him. Conversely, an imprecise signal—by obscuring the latent state, e.g., through vague or confusing content—increases the probability that the agent (incorrectly) infers the candidate is winning and thus backs him. The biased planner is therefore incentivized to *decrease* an agent’s signal precision when it believes the candidate is losing. Any degree of tailoring, in either direction, requires research into how the agent best assimilates content and thus incurs a cost: even reducing precision is costly, since the planner must know, e.g., that an agent best understands focused, brief messages in order to confuse it with long, verbose, detailed articles. The planner selects the precisions so as to maximize the expected number of backers net of these costs.

B SOCIAL LEARNING DYNAMICS

We first summarize the sequence of events at step i :

1. The planner selects q_i as a function of $\mathcal{H}_i$, and q_i is observed by all agents.
2. Agent i receives a private signal s_i satisfying $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i selects the action a_i that maximizes its expected utility, conditioning on the observable history and its private signal.
4. All other agents and the planner observe a_i , and the public belief b_{i+1} at the end of step i is updated accordingly.

By (1), q_i is a deterministic function of $\mathcal{H}_i$; hence conditioning a random variable jointly on q_i and $\mathcal{H}_i$ renders the conditioning on q_i redundant. We invoke this fact repeatedly below.

B.1 PRIVATE BELIEF UPDATE

We now derive agent i ’s action a_i . Let $\alpha_i := \frac{b_i}{1-b_i}$ denote the public likelihood ratio implied by the observed actions. Agent i forms the private belief $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$ by combining the public belief with its private signal s_i , and we denote the associated private likelihood ratio by $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$. We compute $\tilde{\alpha}_i$ and $\tilde{b}_i$ as follows.

1. If private signal is $s_i = G$, then

$$\begin{aligned}\tilde{\alpha}_i &= \frac{\tilde{b}_i}{1 - \tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1 - q_i} \frac{b_i}{1 - b_i} \tag{8}\end{aligned}$$

$$= \frac{q_i}{1 - q_i} \alpha_i \tag{9}$$

In deriving Equation (8), we invoke the conditional-independence assumption—that, given ω , each agent's signal is independent of the other signals and of the history—to write $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$, and analogously for $\omega = B$; we additionally use that q_i is a function of $\mathcal{H}_i$.

2. If the private signal is $s_i = B$, the same argument yields

$$\tilde{\alpha}_i = \frac{1 - q_i}{q_i} \alpha_i \tag{10}$$

Let $y(b, q)$ and $z(b, q)$ denote, respectively, the probabilities of realizing signal G and signal B at public belief b and precision q , conditioned on the history. Conditioning on ω and applying the law of total probability yields:

$$\begin{aligned}y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\ &= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\ &= q_i b_i + (1 - q_i)(1 - b_i) = 1 + 2q_i b_i - q_i - b_i \tag{11}\end{aligned}$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \tag{12}$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) yields the private-belief update:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1 - q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \tag{13}$$

Note that if $s_i = G$ then $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i) > 0$, and if $s_i = B$ then $z(b_i, q_i) > 0$; hence the update above is well-defined in every case.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i 's action a_i equals G when $\tilde{b}_i > 0.5$ (equivalently, $\tilde{\alpha}_i > 1$), B when $\tilde{b}_i < 0.5$ (equivalently, $\tilde{\alpha}_i < 1$), and s_i otherwise by the tie-breaking rule. This action, together with the precision q_i , is observed by all peers and induces the updated public belief.

Consider $b_i < 1 - q_i$, so that $\alpha_i < \frac{1 - q_i}{q_i}$. If $s_i = G$, then $\tilde{\alpha}_i < 1$ by Equation (9); if $s_i = B$, then $\tilde{\alpha}_i < \frac{(1 - q_i)^2}{q_i^2} \leq 1$ by Equation (10) (using $q_i \geq 0.5$). Hence $a_i = B$ for every signal realization when $b_i < 1 - q_i$. A symmetric argument gives $a_i = G$ whenever $b_i > q_i$. These are precisely the regimes in which the strength of the public belief dominates the informativeness of the private signal.

Now consider $1 - q_i \leq b_i \leq q_i$. If $s_i = G$, then $\tilde{\alpha}_i \geq 1$ by Equation (8) together with $b_i \geq 1 - q_i$; if $s_i = B$, then $\tilde{\alpha}_i \leq 1$ by Equation (10) together with $b_i \leq q_i$. Combined with the tie-breaking rule, this gives $a_i = s_i$, establishing Equation (2), reproduced below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Consequently, agent i 's action is informative only in the first case of Equation (2), in which it permits future agents to perfectly recover i 's private signal; there the public belief updates exactly as the private belief in Equation (13). In the remaining two cases the action conveys no information about the true state, so the public belief is unchanged. This yields Equation (3), reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \tag{14}$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now show that, for each $a \in \{G, B\}$, the conditional probability $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ depends on $\mathcal{H}_i$ only through b_i (and on q_i). It then follows that

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \quad (15)$$

For the second and third cases of Equation (2), $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ manifestly depends on $\mathcal{H}_i$ only through b_i (and on q_i) for each $a \in \{G, B\}$. We treat the first case next.

$$\begin{aligned} \mathbb{P}(a_i = G | \mathcal{H}_i, q_i) &= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\ &= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\ &= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\ &= b_i q_i + (1 - b_i)(1 - q_i) \end{aligned} \quad (16)$$

The first step applies the law of total probability. The second substitutes b_i for $\mathbb{P}(\omega = w | \mathcal{H}_i, q_i)$, using that q_i is a function of $\mathcal{H}_i$. The third uses $a_i = s_i$ on $1 - q_i \leq b_i \leq q_i$ together with the conditional independence of s_i from the history given ω . The result follows.

This yields the following theorem:

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G | b_i, q_i, a_i\} \quad (17)$$

Equivalently, b_i is a sufficient statistic for the information in $\mathcal{H}_i$ relevant to the update of b_{i+1} .

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G | b_i, q_i\}}{\mathbb{P}\{a_i | b_i, q_i\}}$$

We prove the claim by induction on i .

Base case. For $i = 1$ we have $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. Applying Bayes' theorem,

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G | b_1, q_1\} \mathbb{P}\{a_1 | \omega = G, b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1 | \omega = G, b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G | b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\} \mathbb{P}\{\omega = G | b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G | b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} \end{aligned}$$

where the final step uses that the unconditioned belief in ω equals the prior b_1 .

Inductive step. Assume the claim holds for all $i \leq n$. Then

$$\begin{aligned} b_{n+1} = \mathbb{P}\{\omega = G | \mathcal{H}_n, a_n, q_n\} &= \frac{\mathbb{P}\{\omega = G | \mathcal{H}_n, q_n\} \mathbb{P}\{a_n | \omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | \mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n | \omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | \mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G | \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | b_n, q_n\} \mathbb{P}\{\omega = G | \mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G | b_n, q_n\}}{\mathbb{P}\{a_n | b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n . This completes the induction and the proof. $\square$

We have thus established Markovianity of the public belief by adapting the arguments of Banerjee (1992) and Bikhchandani et al. (1992). The adaptation is necessitated by the heterogeneity in agents' precisions introduced by the central planner.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We establish the claim separately for each case of a_i in Equation (2). For the second case,

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \mathbb{P}(\omega \neq G | b_i, q_i) = 1 - b_i$$

Symmetrically, for the third case of Equation (2),

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \mathbb{P}(\omega \neq B | b_i, q_i) = b_i$$

For the case $1 - q_i \leq b_i \leq q_i$, in which $a_i = s_i$ by Equation (2),

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \tag{18}$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \tag{19}$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1 - q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \tag{20}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three cases yields the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Substituting Equation (4) into $r_A(b, q)$ in Equation (5) yields

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \tag{21}$$

As established in Section 3.2, we may restrict the altruistic planner to precisions $q \in [p, 1]$, since welfare is non-decreasing in precision (Appendix C.9), and $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) implies optimal precision p whenever $p < \max(b, 1 - b)$: if the signal is insufficiently precise to alter the agent's action, the planner expends no resources on it. In the complementary case, concavity and monotonicity of $\beta(\cdot)$ make $\dot{\beta}(\cdot)$ positive and decreasing, so $r_A(b, q)$ is convex in q over this range; hence, if the optimum lies in this range, it is attained at $q = 1$.

We now identify the belief regime in which precision 1 is optimal.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1 - p \leq \frac{\beta(1)}{C}$, this inequality is never satisfied and precision 1 is never optimal. Otherwise it holds precisely for $b \in [\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C}]$. This establishes Theorem 1. $\blacksquare$

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We prove by induction that the expected k -th-stage reward—the planner’s expected utility from the control and action of the k -th agent—is convex in the public belief. The base case is the instantaneous reward Equation (5), which is convex in the public belief.

The primary difficulty is the atypical structure of the public-belief update: although the state space is uncountable, the update from any state is supported on at most two values. To exploit this, we introduce a *decision tree*—the complete binary tree of all trajectories induced by a fixed initial belief and policy—in which each node has two children, one per signal realization of the next agent. The root is the expected instantaneous reward at time 1 (from the first agent’s action), and the k -th level comprises 2^{k-1} nodes, each indexed by a realization of the first $k-1$ signals.

We then show that if a node at level $k-1$ has convex expected reward, both of its children at level k inherit this property. This is the step at which the action’s dependence on the public belief must be handled: under identical precision and identical signal realization, two agents at distinct public beliefs may select opposing actions (see Equation (2)), so standard bounds on the future terms of a Markov process do not apply. Our specific belief update is nonetheless advantageous, as Bayesian updates form a martingale ($\mathbb{E}[b_{i+1}] = b_i$); combining this martingale property with convexity of the instantaneous reward completes the inductive step and hence the proof.

Preliminaries We first establish the following lemma collecting several properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1-b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$, and let Λ_n and Λ_b be the decision trees rooted at n_0 and b_0 , respectively. It suffices to show that Λ_n and Λ_b are reflections of one another, which we do by establishing that their expected rewards, belief-state values, and branching probabilities are pairwise mirror images.

Consider first the instantaneous reward r_A . Since the agents’ action rule Equation (2) is symmetric, $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$, and therefore

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - Cn_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - Cb_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Hence the instantaneous reward is symmetric about 0.5.

Now consider the belief update from n_i under precision q_i , writing $\bar{x}$ and $\underline{x}$ for the beliefs updated from x after observing action G and B , respectively. From Equation (13),

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \underline{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Hence the belief updates are symmetric as well. Combining this with the symmetry of the reward,

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \underline{b}_i) = r(\underline{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus the expected rewards and belief-state values of Λ_n and Λ_b are mirror images.

Finally, consider the branching probabilities $y(b, q) = 1 + 2bq - b - q$ (action G) and $z(b, q) = 1 - y(b, q)$ (action B):

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Hence the branching probabilities of Λ_n and Λ_b are also symmetric, completing the argument.

Extremal values

We show that $V_A^*(0) = 0$; $V_A^*(1) = 0$ then follows from the symmetry about 0.5 established above.

By Equation (3), at $b_0 = 0$ we have $b_0^+ = b_0^- = 0$ for every precision. Thus, irrespective of the signal realizations or control actions, the agents' actions and the public belief are invariant: $b_i = 0$ for all i and, by Equation (2), $a_i = B$ for all i .

From Equation (5), we then have

$$\begin{aligned}
 V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\
 &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\
 &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - C b_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\
 &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right] \tag{22}
 \end{aligned}$$

Thus $V_A^*(0)$ is attained by minimizing $\beta(\pi(0))$ at every i . Since $\beta(\cdot)$ is increasing on $[0.5, 1]$ with minimum $\beta(p) = 0$, setting $q_i = p$ for all i makes Equation 22 yield $V_A^*(0) = 0$, with optimal policy $\pi_A^*(0) = p$.

Symmetry about 0.5 then gives $V_A^*(1) = 0$, completing the proof. $\square$

We now formalize the notion of a *decision tree* for Markovian and general policies. Any policy is represented by a complete binary tree (see figure 3). Each node encodes the system state—the public belief at which the controller transmits the private signal to the agent at that node's level—and each branch encodes the action the agent takes upon receiving its signal. The controller's policy (its precision choice) determines both the action probabilities from each node and the belief resulting from each action; the action probability at a node is the weight of the corresponding branch. A decision tree is thus fully specified by its node belief values and branch weights. Under a deterministic stationary Markovian policy, the precision—and hence each action probability—is a deterministic function of the node's belief alone, independent of the node's level or the path from the root; consequently the entire tree (node beliefs and branch weights) is determined by the initial state. Under an arbitrary (history-dependent) policy, by contrast, the controller's decision at a node may depend on the full path leading to it.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to public belief.

Proof.

We first show that, for fixed precision, the instantaneous reward r_A is convex in the public belief. Using Lemma 7, rewrite r_A as

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave; subtracting it renders r_A convex in x .

Fix arbitrary x_0, m_0, n_0 with $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. Convexity of $V_A^*(\cdot)$ requires $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. Since $V_A^*(x) \geq V_A^\pi(x)$ for every policy π by optimality, it suffices to exhibit a policy $\tilde{\pi}$ with

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \tag{24}$$

Let Λ denote the decision tree of π^* with initial state x_0 .

We define $\tilde{\pi}$ to be the controller policy that, at each node, applies the precision that Λ applies at the counterpart node, where two nodes are *counterparts* if they are reached by the same action sequence from the root (see figure 3). Thus $\tilde{\pi}$ replicates Λ 's precision after any given action history. We denote by Λ_m (respectively Λ_n) the decision tree of $\tilde{\pi}$ with initial state m_0 (n_0).

We now describe $\tilde{\pi}$ on Λ_m and Λ_n , tracking counterpart nodes and branch weights. Fix a single action path λ through the trees by specifying the set $\mathcal{G}$ of epochs at which action G is taken, with $\mathcal{B}$ its complement (see figure 3). In Λ , let x_i denote the state at epoch i obtained from x_0 under π^* along λ ; thus x_i takes one of 2^i

possible values. In Λ_m (respectively Λ_n), let m_i (n_i) denote the public belief obtained from m_0 (n_0) along λ ; the node m_i (n_i) is the counterpart of x_i . By construction $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, and $\tilde{\pi}$ is defined analogously at every node of Λ_m via the counterpart node of Λ . The weight of the G -branch (respectively B -branch) emanating from x_i in Λ is $y(x_i, \pi_A^*(x_i)) (z(x_i, \pi_A^*(x_i)))$; since $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, the weight of the G -branch from m_i in Λ_m is $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$, and analogously for B -branches and for Λ_n . The three trees therefore differ in both their node values and their branch weights. Finally, since π^* —and hence $\tilde{\pi}$ —is deterministic, Equation (13) and Equation (3) make x_i , m_i , and n_i deterministic once the path $\mathcal{G}$ is fixed.

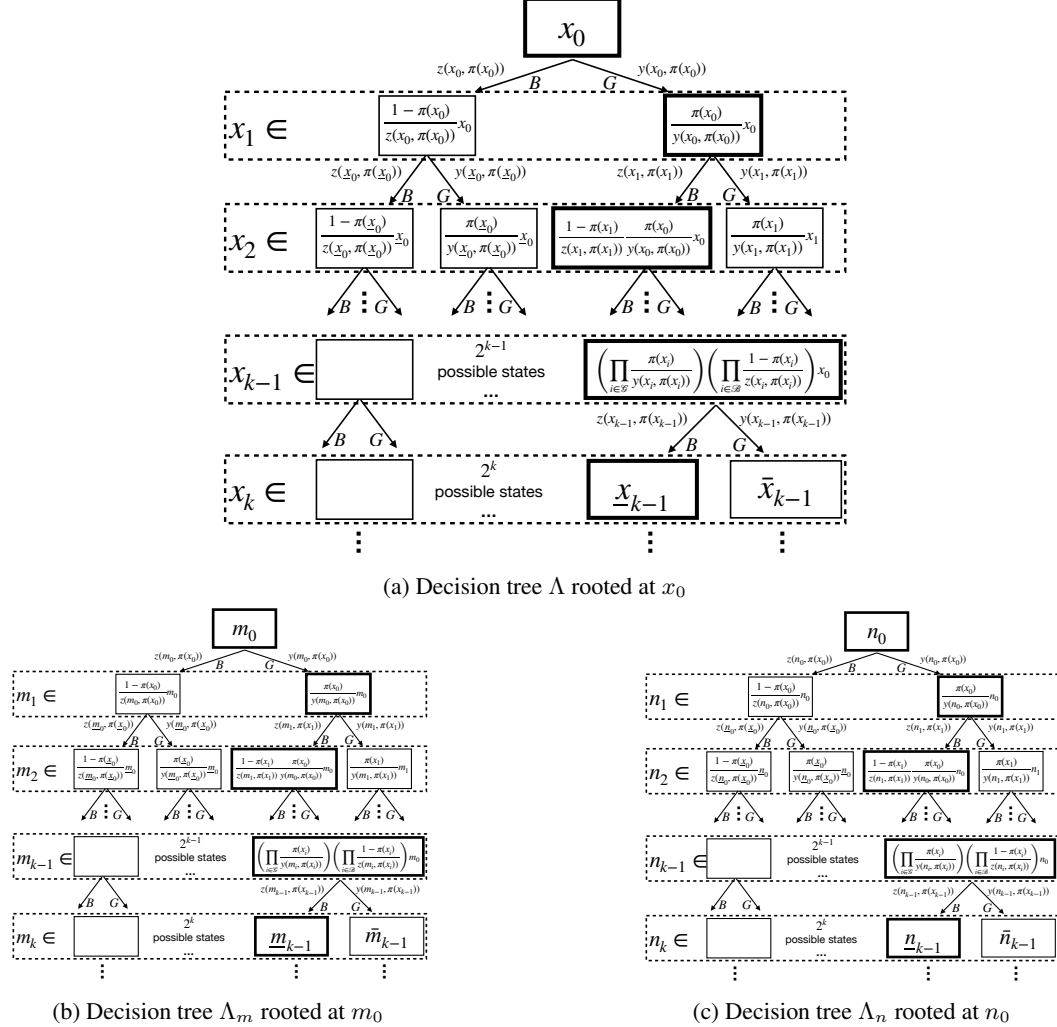


Figure 3: Belief evolution represented as three binary trees rooted at x_0 , m_0 , and n_0 , respectively; level i contains the 2^i states reachable after i epochs. A left branch corresponds to action B and yields the updated belief $\underline{x}_0$ (respectively $\underline{m}_0$, $\underline{n}_0$); a right branch corresponds to action G and yields $\bar{x}_0$ (respectively $\bar{m}_0$, $\bar{n}_0$). Each branch is labeled with its transition probability. The bold states trace an example path λ with $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$; the bold states at each level are mutual counterparts.

Let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ denote the expected instantaneous reward at epoch k starting from b_0 under π , and set $\phi^*(x) = \phi^{\pi_A^*}(x)$. By Fubini's theorem, the value function decomposes as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

Thus (24) is equivalent to

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

which we establish by proving the per-epoch inequality, for every $i = 0, 1, 2, \dots$,

$$\phi_i^*(x_0) \leq t\phi_i^{\tilde{\pi}}(m_0) + (1-t)\phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

summing (27) over i yields (26), hence (24), hence convexity of $V^*(\cdot)$ in the public belief.

Fix a general epoch $i = k$. The term $\phi_{k-1}^{\pi}(x_0)$ comprises 2^{k-1} summands, one per action path in $\{G, B\}^{k-2}$. In particular, the path λ contributes to $\phi_{k-1}^*(x_0)$ the summand

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

Define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$, the probability of traversing λ from x_0 under π_A^* . As before, $\bar{x}_i$ and $\underline{x}_i$ denote the beliefs updated from x_i after action G and B , respectively; equivalently, $\bar{x}_i$ and $\underline{x}_i$ are the belief values of the two level- $(i+1)$ children of the level- i node x_i (see levels $k-1$ and k in figure 3a). The two children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ of x_{k-1} contribute the following pair of terms to $\phi_k^*(x_0)$:

$$P_\lambda^*(x_0) [y(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (29)$$

Analogously, define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$, the probability of traversing λ from m_0 under $\tilde{\pi}$; since $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, this equals $\left(\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)) \right)$. Define $P_\lambda^{\tilde{\pi}}(n_0)$ by substituting $\{n_i\}$ for $\{m_i\}$.

The corresponding terms in $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

$$P_\lambda^*(m_0) [y(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) [y(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (31)$$

We will show that

$$\begin{aligned} P_\lambda^*(x_0) [y(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\ \leq tP_\lambda^*(m_0) [y(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\ + (1-t)P_\lambda^{\tilde{\pi}}(n_0) [y(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \end{aligned} \quad (32)$$

The same combination applies to every sibling pair descending from each level- $(k-1)$ term of the belief tree, yielding an analogous inequality in each case. Summing these inequalities gives (27), and hence (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We focus on the second term of (29) and establish the following two identities:

$$\begin{aligned} \underline{x}_{k-1} &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned} 1 &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \end{aligned} \quad (34)$$

Setting $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, identities (33) and (34) give

$$\underline{x}_{k-1} = \tilde{t}\underline{m}_{k-1} + (1-\tilde{t})\underline{n}_{k-1} \quad (35)$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, we obtain

$$\tilde{t}z(x_{k-1}, \pi_A^*(x_{k-1}))P_\lambda^*(x_0) = tz(m_{k-1}, \pi_A^*(x_{k-1}))P_\lambda^{\tilde{\pi}}(m_0) \quad (36)$$

and, symmetrically,

$$\tilde{t}z(x_{k-1}, \pi_A^*(x_{k-1}))P_\lambda^*(x_0) = (1-t)z(n_{k-1}, \pi_A^*(x_{k-1}))P_\lambda^{\tilde{\pi}}(n_0) \quad (37)$$

Applying these to the second term of (29) yields

$$\begin{aligned} & P_\lambda^*(x_0)z(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\ & \leq P_\lambda^*(x_0)z(x_{k-1}, \pi_A^*(x_{k-1})) \\ & \quad [\tilde{t}r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t})r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\ & = t \left[P_\lambda^{\tilde{\pi}}(m_0)z(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \quad + (1-t) \left[P_\lambda^{\tilde{\pi}}(n_0)z(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (\text{Eqns 36, 37}) \end{aligned} \quad (38)$$

The identical argument applied to the first term of (29) gives

$$\begin{aligned} & P_\lambda^*(x_0)y(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\ & \leq t \left[P_\lambda^{\tilde{\pi}}(m_0)y(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \right] \\ & \quad + (1-t) \left[P_\lambda^{\tilde{\pi}}(n_0)y(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \right] \end{aligned} \quad (39)$$

Summing (38) and (39) yields (32).

It remains to complete the proof of (32)—and thereby the entire proof—by establishing (33) and (34).

Proof of (33)

Iterating the recurrence in Equation (13) and Equation (3), we express $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as products of the per-step multiplicative belief updates applied to x_0 , m_0 , and n_0 , respectively, where each factor is the multiplicative state update induced by a single action:

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \quad (40)$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \quad (41)$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \quad (42)$$

These identities are illustrated in figure 3. Reorganizing the factors and using $x_0 = tm_0 + (1-t)n_0$ relates $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$:

$$\begin{aligned} \underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ & \quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \end{aligned} \quad (43)$$

This proves (33).

Proof of (34)

Consider the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We show that it can be written as $\alpha_0 m_0 + \beta_0$ for coefficients α_0, β_0 that depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$, with the functional form of the dependence fixed by the action sequence along the path in stages 1 through $k-1$. We

proceed by induction backward from the final $((k-1)$ -th) factor. The base case is this final factor, which is either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$ according to the $(k-1)$ -th action:

$$\begin{aligned} y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1})) \end{aligned}$$

Hence the final factor has the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$, where α_{k-1} and β_{k-1} depend only on $\pi_A^*(x_{k-1})$ and whose form is fixed by the $(k-1)$ -th action (which selects $y(\cdot)$ versus $z(\cdot)$).

For the inductive step, assume $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) = \alpha_{j+1}m_{j+1} + \beta_{j+1}$ for coefficients $\alpha_{j+1}, \beta_{j+1}$ depending only on $\{\pi_A^*(x_i)\}_{i > j}$, with form fixed by the actions from $i > j$ to $i = k-1$. We show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) = \alpha_j m_j + \beta_j$, with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ and form fixed by the actions from $i > j-1$ to $i = k-1$.

Define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$. Thus,

$$\begin{aligned} &\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1}) \end{aligned}$$

Thus, from the definitions of $f(\cdot), g(\cdot), y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In both cases, by the inductive hypothesis, $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the actions from $i > j$ to $i = k-1$; hence the claim holds for α_j, β_j , completing the induction.

Now, the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$, which shares the policy and action sequence of the numerator. Since the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the action path, the same argument writes the denominator as $\alpha_0 x_0 + \beta_0$ with identical α_0, β_0 , and likewise for the numerator of the second term in (34). We therefore evaluate the coefficient in (33) as

$$\begin{aligned} &t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ &= \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\ &= \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = tm_0 + (1-t)n_0) \\ &= 1 \end{aligned}$$

Hence (34) holds, completing the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Suppose, for contradiction, that there exists $b \in [0, 1]$ with $\pi_A^0(b) > \pi_A^*(b)$. We construct a policy π for which $V_A^*(b) < V_A^\pi(b)$, contradicting the optimality of $V_A^*(\cdot)$.

Let π apply precision $\pi_A^0(b)$ at the current step from public belief b and the optimal policy thereafter. We expand $V_A^*(b) - V_A^\pi(b)$ into its instantaneous-reward and discounted-future components, writing b^+ and b^- for the beliefs updated after signals G and B under the optimal policy, and b_m^+ and b_m^- for the corresponding updates under the myopic policy.

$$\begin{aligned} V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\ &\quad + \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\ &\quad - \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\ &< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\ &\quad - \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\ &\leq 0 \quad (\text{Theorem 2}) \end{aligned}$$

The first inequality uses that the instantaneous reward under $\pi_A^0(\cdot)$ is maximal, by definition of the myopic optimum.

The final inequality invokes the convexity of $V_A^*(\cdot)$ from Theorem 2: by Equation (13), $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$, and convexity applied to this ordering yields the bound and completes the proof. ■

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ under a policy π that applies precision q at the current step and the optimal policy thereafter; the maximizer q is the optimal precision at public belief b . Since the planner never benefits from precision below the baseline p (Appendix C.9), we restrict $q \in [p, 1]$ without loss of generality.

Let b^+ and b^- denote the belief updates after signals G and B under precision q . The second derivative of $V_A^\pi(b)$ with respect to q , for $q \geq \max(b, 1 - b)$, is

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2 (1 - b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1 - b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

By the convexity of $V_A^*(\cdot)$ and the concavity of $\beta(\cdot)$, this expression is non-negative, so $V_A^\pi(b)$ is convex in q on $[\max(b, 1 - b), 1]$ and is therefore maximized at an endpoint of this interval. On $q \in [0.5, \max(b, 1 - b))$ the derivative of $V_A^\pi(b)$ in q is negative, so the cost-minimizing baseline p is optimal there. Hence the optimal precision belongs to $\{p, \max(b, 1 - b), 1\}$.

The expected value under the latter two candidates is strictly negative, whereas under precision p it equals 0 at $b = 0$ and $b = 1$. Combined with the non-positivity and symmetry of the value function (Lemma 8), this yields a threshold d_A with $\pi_A^*(b) = p$ for $b \in [0, d_A) \cup (1 - d_A, 1]$. Moreover, Theorem 1 together with Lemma 9 gives $d_A \in (0, t_M)$, with t_M as in Theorem 1.

Finally, Theorem 1 and Lemma 9 also yield a threshold $t_A \in [d_A, t_M]$ with $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$, completing the proof. ■

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Substituting the action rule (2) into the instantaneous reward (6) yields the piecewise form:

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Observe that $z(b, q) = b + q - 2bq$ is increasing in q for $b < 0.5$ and decreasing in q for $b > 0.5$.

At $b = 0.5$, $z(0.5, q) = 0.5$ independently of q . Moreover, since $q \geq 0.5$ always holds, $q \geq \max(b, 1 - b)$ at $b = 0.5$, placing us in the first case of (44); there the second term of $r_B(b, q)$ is constant in q , so the planner maximizes the first term alone, attaining the optimum at $q = p$.

We now establish a sequence of lemmas that support the proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable in q everywhere except at $q = p$, where its derivative (for $q \neq p$) is

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) \\ -C(1 - 2b) \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) \end{cases} \quad \begin{matrix} q \geq \max(b, 1 - b) \\ \\ \text{otherwise} \end{matrix} \quad (45)$$

In the former case, the second derivative in q (for $q \neq p$) is

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Since $\beta(\cdot)$ is concave, this is non-negative, so $r_B(b, q)$ is convex in q . The maximum is therefore attained either at an endpoint $\{\max(b, 1 - b), 1\}$ or at the non-differentiability point $q = p$. $\square$

Lemma 11.

For $b \in [0, 1 - p]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Since $p > 0.5$, we have $\max(b, 1 - b) = 1 - b > p$ for $b \in [0, 1 - p]$.

First, if $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$, then by (44) the reward is maximized by minimizing $|q - p|$, so the optimal precision below $\max(b, 1 - b)$ is p .

Next, if $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$, then Lemma 10 gives $\pi_B^0(b) \in \{1 - b, p, 1\}$, and it remains to exclude $\pi_B^0(b) = 1$. Comparing $r_B(b, 1)$ and $r_B(b, 1 - b)$ —both in the first case of (44)—the first reward term is larger at $1 - b$ than at 1 because $\beta(\cdot)$ is increasing and $p < 1 - b \leq 1$. Since $z(b, q)$ is increasing in q for $b < 0.5$, we obtain $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$, excluding precision 1 and completing the proof. $\square$

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. For any $q \neq p$ the reward is strictly negative, owing to the term $-\beta(|q - p|)$. For $b \in (p, 1]$ and $q = p$ we have $p < \max(b, 1 - b) = b$, placing us in the third case of (44) with reward 0. Hence $q = p$ is optimal. $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. If $p = 0.5$, the interval $[1 - p, 0.5]$ reduces to $\{0.5\}$, at which $p = 1 - b$ and thus $\pi_B^0(b) = p = 1 - b = 0.5$. We treat $p > 0.5 \geq b$ for the remainder.

For $b \in [1 - p, 0.5]$ we have $\max(b, 1 - b) = 1 - b \leq p$, so by the first case of (44) precision p yields $r_B(b, p) = -Cz(b, p)$. We distinguish $q < \max(b, 1 - b)$ from $q \geq \max(b, 1 - b)$.

If $q < \max(b, 1 - b) = 1 - b$, the second case of (44) gives reward $-\beta(|q - p|) - C$. Since $z(b, q)$ is increasing in q for $b < 0.5$ with range $[z(b, 0.5), z(b, 1)] = [0.5, 1 - b]$, we have $z(b, q) < 1$ for all q ; hence this reward is strictly below $r_B(b, p)$, excluding the case.

If $q \geq \max(b, 1 - b)$, Lemma 10 restricts the candidates to $\{1 - b, p, 1\}$. To exclude precision 1, compare it with p : the first reward term in (44) is larger at p , and since $z(b, q)$ is increasing in q for $b < 0.5$ the second term is also larger at p . Hence precision 1 is excluded, completing the proof. $\square$

We now define the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

By definition, every optimal policy is ϵ -optimal for all $\epsilon > 0$; in particular, $\pi_B^0(b)$ is ϵ -optimal whenever it exists.

Accordingly, let $\pi_B^{\epsilon, 0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or

$$2. \pi_B^{\epsilon,0}(b) = b - \epsilon$$

Proof. For $b \in (0.5, p]$ we have $\max(b, 1 - b) = b \leq p$. We distinguish $q \geq \max(b, 1 - b)$ from $q < \max(b, 1 - b)$.

In the first case, Lemma 10 gives $\pi_B^0(b) \in \{b, p, 1\}$. Comparing precision b with p : the first reward term in (44) is larger at p , and since $z(b, q)$ is decreasing in q for $b > 0.5$ the second term is also larger at p . Hence precision b is excluded, yielding the first claim.

In the second case, $q < \max(b, 1 - b) = b$, so (i) $q \in [0.5, b)$ (as $q \geq 0.5$) and (ii) the third case of (44) applies, giving reward $-\beta(|q - p|)$. Since $0.5 < b \leq p$, this reward is increasing in $q \in [0.5, b)$, yielding the second claim. $\square$

Remark 3. As noted in Puterman (1990) (Section 2.3.1), an optimal policy may fail to exist over an infinite action space when the supremum is not attained. In such cases we instead seek ϵ -optimal policies—policies whose reward is within $\epsilon > 0$ of the supremum. This situation arises, in particular, in case 2 of Lemma 14.

Proof of Theorem 4 Using $p \geq 0.5$, we partition $[0, 1]$ into four cases: $b \in [0, 1 - p)$, $b \in [1 - p, 0.5)$, $b \in (0.5, p]$, and $b \in (p, 1]$. By Lemmas 11–14, the respective candidate policies are $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$.

Case 1: $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

We establish a threshold t_1 with $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

By (44), the rewards under the two candidate policies are

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1 - b) &= -\beta(1 - b - p) - C(1 - 2b(1 - b)) \end{aligned}$$

The derivative of $r_B(b, 1 - b)$ in b is

$$\frac{\partial}{\partial b} r_B(b, 1 - b) = \dot{\beta}(1 - p - b) + 2C(1 - 2b)$$

which is positive since $\beta(\cdot)$ is increasing and $b < 1 - p \leq 0.5$. Evaluating at the endpoints $b = 0$ and $b = 1 - p$,

$$\begin{aligned} r_B(0, 1) &= -\beta(1 - p) - C < r_B(b, p) \\ r_B(1 - p, p) &= -C(1 - 2p(1 - p)) > r_B(b, p) \end{aligned}$$

Thus $r_B(b, 1 - b)$ is increasing on $[0, 1 - p)$ with $r_B(b, 1 - b) < r_B(b, p)$ at $b = 0$ and the reverse at $b = 1 - p$. Since both reward functions are continuous ($\beta(\cdot)$ being continuous), there exists $t_1 \in (0, 1 - p)$ with $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

Case 2: $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{1 - b, p\}$ (Lemma 13)

If $p = 0.5$, the interval is $\{0.5\}$, at which $1 - b = p$; the candidates coincide and $\pi_B^0(b) = p$. For the remainder we take $p > 0.5$.

We establish thresholds t_2, t_3 with $\pi_B^0(b) = 1 - b$ for $b \in [t_2, t_3)$ and $\pi_B^0(b) = p$ for $b \in [1 - p, t_2] \cup [t_3, 0.5]$.

By (44), the rewards under the candidate policies are

$$\begin{aligned} r_B(b, 1 - b) &= -\beta(p - 1 + b) - Cz(b, 1 - b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

The condition for precision $1 - b$ to be optimal is therefore

$$\begin{aligned} r_B(b, 1 - b) &\geq r_B(b, p) \\ \beta(p - 1 + b) &\leq C[z(b, p) - z(b, 1 - b)] \\ \beta(p - 1 + b) &\leq C(p - 1 + b)(1 - 2b) \end{aligned} \tag{46}$$

The left-hand side of (46) is a concave, increasing function of b on $[1 - p, 0.5]$; the right-hand side is a concave parabola that increases on the first half of $[1 - p, 0.5]$ and decreases on the second half.

Partition the interval into $[1 - p, 0.75 - 0.5p]$ and $[0.75 - 0.5p, 0.5]$. On the first subinterval both sides are increasing and concave and the right-hand side is a parabola, so they intersect at most twice (one intersection being $b = 1 - p$). On the second subinterval the left-hand side is increasing while the right-hand side is decreasing, so they intersect at most once. Denote the two possible intersections for $b > 1 - p$ by $t_2 \leq t_3$.

It remains to determine the optimum on each of $(1 - p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $r_B(b, p) > r_B(b, 1 - b)$ (since $p > 0.5$), so $\pi_B^0(b) = p$ on $(t_3, 0.5)$; consequently $\pi_B^0(b) = 1 - b$ on (t_2, t_3) and $\pi_B^0(b) = p$ on $(1 - p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b - \epsilon\}$ (Lemma 14)

We establish $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b - \epsilon$, respectively.

By (44), the rewards under the candidate policies are

$$\begin{aligned} r_B(b, 1) &= -\beta(1 - p) - C(1 - b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b - \epsilon) &= -\beta(p - b + \epsilon) \end{aligned}$$

At $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0, so $b - \epsilon$ is optimal on some interval $(t_5, p]$ with $0.5 \leq t_5 < p$.

Since $r_B(b, 1)$ and $r_B(b, p)$ are affine in b while $r_B(b, b - \epsilon)$ is convex in b (by concavity of β), the differences $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and hence have convex superlevel sets. The set on which both are positive—i.e., on which $b - \epsilon$ is optimal—is therefore convex, so $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

It remains to compare precisions 1 and p on $(0.5, t_5]$, which we do via their values and derivatives at $b = 0.5$.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Since $r_B(0.5, 1) < r_B(0.5, p)$, there is an interval $(0.5, t_4)$ with $t_4 \leq t_5$ on which $\pi_B^0(b) = p$. As both rewards are affine in b , their ordering changes at most once; hence, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This is immediate from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. By (44), the derivative of the biased reward in b is

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Since $q \geq 0.5$ and $z(b, q) \in [0, 1]$, $r_B(b, q)$ is non-decreasing in b .

Next, the Bayesian belief updates form a martingale: for every fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. We verify this in two cases. If $q_i < \max(b_i, 1 - b_i)$, then (3) gives $b_{i+1} = b_i$, so the claim holds trivially. If $q_i \geq \max(b_i, 1 - b_i)$, then (13) supplies the distribution of b_{i+1} , from which the same identity follows.

Monotonicity of $V_B^*(\cdot)$ then follows from Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first prove that the optimal value function is continuous in the following lemma:

Lemma 16. Biased value function continuity

$V_B^*(\cdot)$ is continuous w.r.t. b .

Proof. Consider first the one-stage value function $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. By Theorem 4, $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$, so we may write

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1 - b, 1, b - \epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

We now show $V_B^0(\cdot)$ is continuous on $[0, 1]$. Consider first $b \leq 0.5$. By Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$; in either case the instantaneous reward is independent of ϵ , so the supremum in (48) may be dropped, leaving $V_B^0(b) = \max_{q \in \{1-b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ lies in the first case of (44) and is thus continuous in b , while $r_B(b, p)$ has a single discontinuity at $b = 1 - p$, where it transitions from the second to the first case of (44). At $b = 1 - p$, Theorem 4 shows the myopic optimum switches from $1 - b$ to p ; both lie in the first case of (44) and, since $p = 1 - b$ there, the controls coincide. Hence the one-sided limits agree with the optimal value and $V_B^0(\cdot)$ is continuous on $[0, 0.5]$.

Consider next $b \geq 0.5$, for which Theorem 4 gives candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained at $q = b - \epsilon$, the third case of (44) gives $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$; since $b - \epsilon$ is selected only for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Interchanging the maximum and supremum in (48) then gives $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. Here $r_B(b, 1)$ lies in the first case of (44) and is continuous, and $-\beta(|p - b|)$ is continuous by continuity of $\beta(\cdot)$. For $r_B(b, p)$ with $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third for $b > p$, so $r_B(b, p)$ is potentially discontinuous at $b = p$; it thus remains to verify continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, Theorem 4 gives $\pi_B^{\epsilon, 0}(p) = b - \epsilon$, so the inner maximum of (48) is attained at $q = b - \epsilon$ and, by the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Likewise, by Theorem 4, for each $\epsilon > 0$ there is a $t_5 < p$ such that the inner maximum is attained at $q = b - \epsilon$ on $[t_5, p]$, where $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$; hence $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimum is $q = p$, giving $V_B^0(b) = 0$ by the third case of (44). This establishes continuity at $b = p$ and hence on all of $[0, 1]$.

Theorem 4.2 of Hinderer (2005) then yields continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ in the public belief and the compactness of the control set $[0.5, 1]$, one might expect an optimal policy to exist at every public belief. This inference fails, however, due to an important subtlety: although $V_B^*(\cdot)$ is continuous in the public belief, it need not be continuous in the precision. Concretely, b_{i+1} depends on both b_i and q_i (see Equation (3)), so $\mathbb{E}[V_B^*(b_{i+1})]$ may be written as $\phi(b_i, q_i)$; since b_{i+1} need not be continuous in q_i , neither need $\phi(\cdot)$. Consequently, there may exist belief regimes in which no policy attains the optimal value. We show below that such regimes indeed exist.

Proof of Theorem 5 We now invoke Lemma 16 to prove Theorem 5. As before, $z(b, q) = b + q - 2bq$ is the probability that an agent receives signal B at precision q and public belief b , and $y(b, q) = 1 - z(b, q)$ that it receives signal G . Abusing notation, let $V_B(b, q)$ denote the expected utility of starting at public belief b , applying precision q at the current step, and following the optimal policy $\pi_B^*(\cdot)$ thereafter. By (13), (3), and (44),

$$V_B(b, q) = \begin{cases} \begin{aligned} & -\beta(|q - p|) - Cz(b, q) \\ & + \delta y(b, q) V_B^*(f(b, q, G)) \\ & + \delta z(b, q) V_B^*(f(b, q, B)) \end{aligned} & q \geq \max(b, 1 - b) \\ \begin{aligned} & -\beta(|q - p|) \\ & - C\mathbf{1}(b < 0.5) \\ & + \delta V_B^*(b) \end{aligned} & \text{otherwise} \end{cases} \quad (49)$$

In the second case of (49), where $q < \max(b, 1 - b)$, the only precision-dependent term is $-\beta(|q - p|)$, maximized by taking q as close to p as possible. Hence, whenever the optimal precision lies below $\max(b, 1 - b)$, it is as close to p as feasible.

We treat four cases: (1) $b \in [0, 1 - p)$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p)$, claims (A) and (B)

We establish a threshold t_1 with $\pi_B^*(b) = p$ for $b \leq t_1$. If $\pi^*(b) < \max(b, 1 - b)$, we are in the second case of Equation (49), whose sole precision-dependent term $-\beta(|q - p|)$ is maximized by minimizing $|q - p|$. Since $p \in [0.5, 1]$ implies $p < \max(b, 1 - b)$ for $b \in [0, 1 - p)$, it follows that on this interval $\pi_B^*(b) < \max(b, 1 - b)$ forces $\pi_B^*(b) = p$.

Now evaluate the value function at $b = 0$. There the state is known to be B , so no signal—at any precision—moves the public belief, as seen by substituting $b = 0$ into (13). Moreover, at $b = 0$, $q \geq \max(b, 1 - b)$ forces $q = 1$, while (as above) $q < \max(b, 1 - b)$ gives optimal precision $q = p$. Comparing these two candidates via Equation (49),

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Since $V_B(0, 1) < V_B(0, p)$, we have $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$, so Equation (51) gives $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15 this is a lower bound on the value function at every public belief; combined with $V_B^*(b) \leq 0$, it yields $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now exhibit an interval $[0, \epsilon]$, $\epsilon > 0$, on which the optimum of the first case of Equation (49) is strictly below that of the second case; there the optimal control lies in the second case, so $\pi_B^*(b) < \max(b, 1-b)$, and the preceding argument then gives $\pi_B^*(b) = p$, completing the proof.

Let $g(\cdot)$ denote the maximum of the first-case right-hand side of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q-p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, Equation (13) makes $f(b, q, G)$ and $f(b, q, B)$ continuous in q ; hence the maximand in Equation (52) is continuous in q over the compact set $[\max(b, 1-b), 1]$, and the maximum is attained.

We now take $\epsilon \rightarrow 0$ in $g(\epsilon)$. Since $q \in [\max(\epsilon, 1-\epsilon), 1]$, $\epsilon \rightarrow 0$ forces $q \rightarrow 1$ inside the maximum in Equation (52), and we treat the four terms in turn. As $q \rightarrow 1$: (i) by continuity of $\beta(\cdot)$, $\beta(|q-p|) \rightarrow \beta(1-p)$; (ii) by continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$ and $y(\epsilon, q) \rightarrow 0$, so the second term tends to C ; (iii) since $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term tends to 0; and (iv) since $z(\epsilon, q) \rightarrow 1$, $f(\epsilon, q, B) \rightarrow 0$, and $V_B^*(\cdot)$ is continuous (Lemma 16), the last term tends to $\delta V_B^*(0)$. Therefore

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1-p) - C + \delta V_B^*(0) \quad (53)$$

By contrast, applying precision p gives, from the second case of Equation (49),

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p-p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality uses the continuity of $V_B^*(\cdot)$ from Lemma 16.

Since Equation (53) is strictly less than Equation (55), there exists an interval $[0, \epsilon]$, $\epsilon > 0$, on which the first-case optimum of Equation (49) lies below the second-case optimum. The result follows.

Case 2: $b \in [1-p, 0.5]$, Claim (C)

We show that $\pi^*(b_i) \geq 1-b_i$ for $b_i \in [1-p, 0.5]$, establishing claim (C). Consider a stationary policy $\pi(\cdot)$ with $\pi(b_i) < 1-b_i$. By Equation (3) the public belief is then invariant, so the same control is applied at every step, and by Equation (2) every agent selects action B . Using $1-b_i \leq p$ on this interval, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, Equation (6) gives the expected total discounted utility under this policy:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Hence this policy attains $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1-\delta}$ —the discounted cost of every agent selecting B under the persistent precision $\pi(b_i)$. Since $\pi(b_i) < 1-b_i \leq p$, this is strictly below the value-function lower bound $\frac{-C}{1-\delta}$ established via Lemma 15 in Case 1. Therefore, if an optimal policy exists here, it satisfies $\pi_B^*(b_i) \geq 1-b_i$.

The remaining candidates are $q_i \in [1-b_i, 1]$. By Bellman's principle,

$$V_B^*(b_i) = \sup_{q_i \in [1-b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Since $q_i \geq \max(b_i, 1-b_i)$ and $b \in [1-p, 0.5] \subset (0, 1)$, Equation (13) and Equation (3) give $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1-q_i)}{z(b_i, q_i)}$, both continuous in q_i . As $V_B^*(\cdot)$ is continuous (Lemma 16), the maximand in Equation (56) is continuous in q_i , so its supremum over the compact set $q_i \in [1-b, 1]$ is attained. An optimal policy therefore exists, with $\pi_B^*(b_i) \geq 1-b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, the assumption $q \in [0.5, 1]$ gives $q \geq b$, placing us in the first case of Equation (49). There $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous in q , so the supremum of $V(b, q)$ over the compact set $q \in [0.5, 1]$ is attained and an optimal policy exists at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ with $\pi'(b) < b$. By Equation (3) and Equation (2), the public belief is then invariant, every agent selects action G , and the same control is applied at every step. Using $b_i \leq p$, $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, Equation (6) gives the total expected discounted utility:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus this policy attains $V_B^{\pi'}(b_i) = \frac{-\beta(p-\pi'(b_i))}{1-\delta}$, which is increasing in $\pi'(b_i)$; hence its supremum over $\pi'(b_i) \in (0.5, b)$ is not attained.

We distinguish two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p-\pi'(b_i))}{1-\delta}$, and (2) its complement. In case (1), the argument used for $b = 0.5$ shows the supremum is attained and an optimal policy exists with $\pi_B^*(b_i) \geq b_i$. In case (2), no optimal policy exists; instead, for sufficiently small $\epsilon > 0$ there is an ϵ -optimal policy with $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

It remains to exhibit $t_2 \in (0.5, p)$ such that case (2) holds for $b_i \in (t_2, p]$, yielding claim (F). Define $m(\cdot)$ and $n(\cdot)$ as the value from precisions below b and at least b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b]} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking $b \rightarrow p$ in $m(\cdot)$ gives

$$\lim_{b \rightarrow p} m(b) = 0 \quad (57)$$

For $q \geq b$ we are in the first case of Equation (49), so $n(b)$ expands as

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q) V_B^*(b, q) \\ &\quad + \delta z(b, q) V_B^*(b, q)] \end{aligned} \quad (58)$$

We claim that $n(b) \leq -C(1 - p)$ for $b \in (0.5, p]$ and $q \geq b$. Since $p < 1$ is fixed, this and Equation (57) give $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, establishing an open interval $(t_2, p]$ on which no optimal policy exists and thereby claim (F). To prove the claim, note that every term of Equation (58) is at most 0, so it suffices to bound one of them; take the second, $-Cz(b, q)$. Since $z(b, q) = q(1 - 2b) + b$ is decreasing in q (as $b > 0.5$), we have $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$, the last inequality using $b \in (0.5, p]$. Hence $-Cz(b, q) \leq -C(1 - p)$, and the claim follows since $p < 1$.

Together with Equation (57), this establishes an open interval (t_2, p) on which no optimal policy exists, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider the stationary policy $\pi(\cdot)$ with $\pi(b) = p$. Since $b > p$, Equation (3) and Equation (2) imply the public belief is invariant, every agent selects action G , and the same control is applied at every step. By Equation (6),

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

so $V_B^\pi(b) = 0$. As 0 is also an upper bound on the optimal value function for every b , we conclude $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. The one-step welfare $-C \min(b_i, 1 - b_i, \pi(b_i))$ is clearly non-decreasing in $\pi(b_i)$.

Moreover, social welfare is convex in the public belief by the same argument as in the proof of Theorem 2, since welfare coincides with the altruistic value function in the special case $\beta(\cdot) = 0$.

The argument of Lemma 9 (Appendix C.4) then shows that increasing precision increases welfare. □

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

We show that, for the altruistic planner, the problem analyzed above is equivalent to one with two agent types of opposing preferences.

Suppose there are two agent types, G and B , with agent i 's type denoted t_i . A type- G agent receives unit reward when $a_i = \omega$ and zero otherwise; a type- B agent receives unit reward when $a_i \neq \omega$ and zero otherwise. Assume every agent knows its own type and those of all others, with all other model components unchanged.

Agent i 's private-belief update is unchanged, as it depends only on the signal structure $\mathbb{P}(s_i = \omega) = q_i$:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_i q_i - b_i - q_i} b_i & s_i = G \\ \frac{1-q}{b_i + q_i - 2b_i q_i} b_i & s_i = B \end{cases} \quad (59)$$

By Equation (59), agent i 's action is as follows, with the superscript denoting the type of agent i and $!$ denoting negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Since types are common knowledge, all agents perfectly infer agent i 's private signal when $q_i \geq \max(b_i, 1 - b_i)$; when $q_i < \max(b_i, 1 - b_i)$, agent i 's action is uninformative.

Consequently, after observing agent i 's type t_i , precision q_i , and action a_i , the public belief updates as

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this setting the altruistic planner seeks to induce type- G agents to take action ω and type- B agents to take action $!\omega$, each of which aligns with the respective agent's utility.

The planner's instantaneous reward is therefore, with the superscript denoting the type of agent i :

$$\begin{aligned} r^G(b_i, q_i) &= -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega) \\ &= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases} \\ r^B(b_i, q_i) &= -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega) \\ &= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases} \end{aligned}$$

These two rewards are identical and coincide with the reward in Equation (5), establishing the equivalence of the altruistic planner's problem under heterogeneous but observable types. ■

E EXPERIMENTS

E.1 EXPERIMENTAL SETUP

To instantiate the controlled social-learning setting, we consider a sequence of agents deciding whether to purchase a new car model, conditioning on the observed history of prior decisions and on an advertisement whose precision is selected by the planner. LLMs occupy three roles: (1) the agents, (2) the planner, and (3) the oracle. Each agent is assigned a profile encoding demographic attributes together with the agent's intended car use cases and preferences. At initialization, the offering is set to either the good or the bad car. The information sets and objectives of the planner and agents are exactly those of the model in Section 3.

The oracle serves two functions: (1) it generates a generic message that realizes the baseline-precision signal absent planner intervention, and (2) given a planner-chosen precision, it generates a message realizing that precision, tailored to the target agent's profile.

The oracle is supplied with a per-car fact sheet (safety rating, customer satisfaction, etc.) and the agent profiles; together these enable generation of a tailored message at the planner-prescribed precision. Further details of the setup and prompts appear in Appendix E.2.

Each agent makes a binary decision conditioned on the history of prior decisions and its private signal, with the objective of matching the true state.

The planner conditions on the same decision history and selects a precision for each agent’s message; it is additionally given the precision of a generic, untailored message that it may send at no cost. The planner’s objective is to induce correct decisions in the altruistic case and to induce purchases in the biased case.

We benchmark the LLM simulations against analytic solutions computed with JuliaPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

All experiments use the LG AI EXAONE Deep 32B model (LG AI Research, 2025). We solve the model analytically and run the LLM simulations over the parameter grid in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	{0.1, 0.3, 0.5, 0.7, 0.9}
p	Baseline signal precision	{0.6, 0.7, 0.8, 0.9}
δ	Discount Factor	{0, 0.25, 0.5, 0.75, 1}

Table 1: Instance parameters and the value grid swept in the experiments.

The car fact sheets were constructed so that the good car is dominant for every profile and the bad car is dominated for every profile. The fact sheets for the GOOD and BAD cars are reproduced below.

```
fact_sheet["GOOD"] = [
    "EPA-rated range: 290 miles.",
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",
    "8-year / 100,000-mile battery warranty.",
    "NHTSA 5-Star overall safety rating.",
    "Cold-soak -5 °F keeps 93 %% range.",
    "Cools cabin 100→72 °F in 3 min.",
    "Cargo volume 26 ft³ total.",
    "Tow's 1 500 lb with factory hitch.",
    "Average 1.2 service visits first 3 yrs.",
    "92 %% owners would buy again (Apr-25).",
    "Qualifies for federal credit.",
    "Quarterly OTA updates incl. battery pre-heat."
]
fact_sheet["BAD"] = [
    "EPA-rated range: 155 mi (combined).",
    "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",
    "Battery warranty: 3 yr / 36 000 mi, first owner only.",
    "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
    "Cold-soak 15 °F retains 58 %% range; cabin heat cuts range 20%.",
    "Cabin cools 105 °F → 72 °F in 11 min.",
    "Cargo volume 17 ft³; rear seats fixed (no fold-flat).",
    "No factory towing approval.",
    "Average 3.8 service visits per year (battery-alert issues).",
    'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
    "Not eligible for federal tax credit.",
    "OTA updates limited to infotainment."
]
```

Profiles were generated programmatically, each specifying the agent’s name, age, city of residence, interests, intended use case for the car (e.g., commuting or road trips), and principal concerns (e.g., fuel efficiency or winter driving).

E.2.2 PROMPTS

Agent LLM Prompt:

```

1782 sys = (
1783     f"You are deciding whether to buy a car. "
1784     f"Your profile is {profile}"
1785     f"You start believing there's a 50% chance the car is GOOD. "
1786     f"You have two information sources: "
1787     f"1. Previous agents' decisions "
1788     f"2. An advisor's message about the car "
1789     f"Your goal is to maximize your reward "
1790     f"i.e. BUY if the car is GOOD and "
1791     f"NOT BUY if the car is BAD. "
1792     f"If you BUY a GOOD car, you GAIN ${C}. "
1793     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1794     f"If you BUY a BAD car, you LOSE ${C}. "
1795     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1796     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1797     f"decision = exactly 'BUY' or 'NOT BUY'."
1798 )
1799 user = (
1800     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1801     f"ADVISOR MESSAGE:\n{message}\n\n"
1802     f"TASK: Use the previous decisions and the advisor message. "
1803     f"Decide BUY or NOT BUY. "
1804     f"Update your belief about car quality (0.0-1.0). "\n"
1805     f'JSON response:{{"belief": <number>,"decision": "<BUY or NOT BUY>"}}'
1806 )
1807
1808 Planner LLM Prompt:
1809
1810 sys = (
1811     f"You are providing messages to potential car buyers. "
1812     f"Each potential car buyer (or agent) has the same information. "
1813     f"Your goal is to get agents to {objective}."
1814     f"Each decision according to your objective, earns you {C} dollars. "
1815     f"You and the agents have the same information. "
1816     f"Initially there is a 50% chance the car is GOOD. "
1817     f"The agents' aim to BUY if the car is GOOD "
1818     f"and NOT BUY if the car is BAD. "
1819     f"You can send a baseline message for free "
1820     f"or tailor the message to the agent at a cost. "
1821     f"Tailoring costs {k} dollars times the change in precision. "
1822     f"(Message precision is the probability (0.5 to 1.0) "
1823     f"that the message leaves the buyer with the correct impression. "
1824     f"For example, if the car is likely GOOD: "
1825     f"a high precision message will likely suggest BUY "
1826     f"and a low precision message will likely suggest NOT BUY. "
1827     f"You may choose to increase or decrease precision. "
1828     f"Agents decide based on your message and the decision history. "
1829     f"They update beliefs by observing previous decisions. "
1830     f"IMPORTANT: Thus your objective is to maximize:"
1831     f"{C} * (# of agents who choose according to your objective) "
1832     f"- sum of ({d}^i*{k}*abs(message_precision - baseline_precision))\n"
1833     f"where {d} is your discount factor for future cost. "
1834 )
1835 user = (
1836     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1837     f"DECISION HISTORY:\n{history_txt}\n"
1838     f"TASK:"
1839     f"Based on the history and your objective, "
1840     f"choose your desired precision.\n"
1841     f'JSON response: {{ "message_precision": <number> }}\n'
1842 )
1843
1844 Oracle LLM Prompt:
1845
1846 sys = (

```

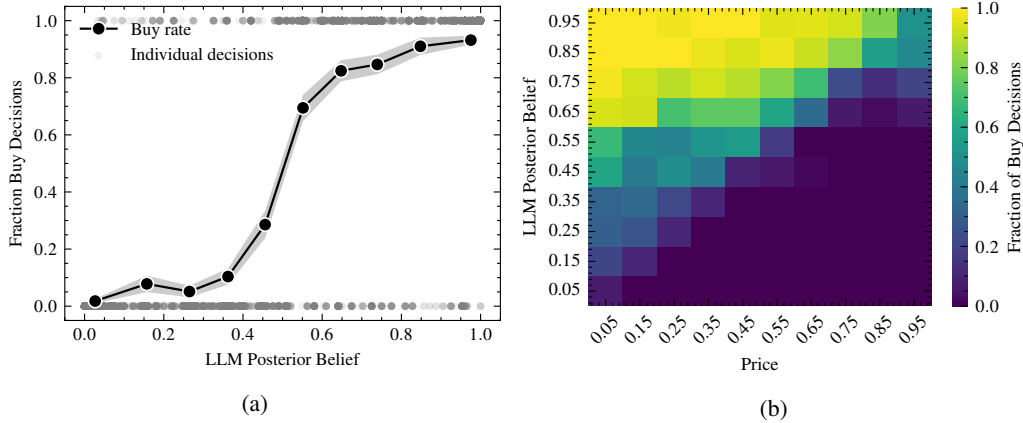
```

1836     f"You will be given "
1837     f"1. an agent profile, "
1838     f"2. information about a car, "
1839     f"3. the true quality of the car, "
1840     f"4. a baseline message and the 'precision' of that message, and "
1841     f"5. a desired precision for the message you output. "
1842     f"(Message precision is the probability (0.5 to 1.0) "
1843     f"that the message leaves the buyer with the correct impression. "
1844     f"For example, if the car is likely GOOD: "
1845     f"a high precision message will likely suggest BUY "
1846     f"and a low precision message will likely suggest NOT BUY. "
1847     f"IMPORTANT: Output a message with the requested precision. "
1848     f"IMPORTANT: The message should be tailored to the profile. "
1849 )
1850 user = (
1851     f"AGENT PROFILE:{profile}\n"
1852     f"TRUE CAR QUALITY: {omega}\n\n"
1853     f"CAR FACTS: {car_facts}\n"
1854     f"BASELINE MESSAGE:{baseline_message}}\n"
1855     f"BASELINE MESSAGE precision: {p}\n"
1856     f"DESIRED MESSAGE precision: {q}\n"
1857     f"IMPORTANT: Output a message with the requested precision. "
1858     f"TASK: Write a message (MAX 100 words) with the desired precision."
1859     f'JSON response: {{"message": "<message>"}}'
1860 )

```

1858 E.3 VALIDATION OF LLM BELIEFS AND ORACLE

1859 We report additional experiments validating the LLMs' self-reported beliefs (cf. figure 1b) and the oracle's performance. Here the LLMs are treated as rational, utility-maximizing—though not necessarily Bayesian—agents. The objectives are (1) to establish that the model's self-reported posterior is the belief that actually governs its choices in state-contingent decision problems, and (2) to verify that the oracle's generated messages realize the requested precision.



1879 Figure 4: Empirical rate at which the LLM selects a ‘buy’ decision, as a function of (a) its self-reported posterior belief, and (b) the joint pair of its self-reported posterior and the purchase price.

1882 First, we supply the LLM with a prior and a new signal, elicit its self-reported posterior belief, and then request a binary “buy/not-buy” decision that yields unit reward if and only if the true state is good. A rational agent selects “buy” precisely when its subjective belief exceeds 0.5. As shown in figure 4a, the LLM’s decisions align sharply with this threshold, selecting “buy” almost exclusively when its reported posterior exceeds 0.5 and “not buy” otherwise. Next, we elicit the model’s willingness to pay for a state-contingent payoff. For a range of prices c , we ask whether the agent would pay c in exchange for unit utility conditional on the state being good. A rational agent with belief p accepts precisely when $c < p$. Figure 4b plots the fraction of “buy” decisions over posterior–price pairs and reveals a sharp separation: acceptance is high for $c < p$ and low for $c > p$, exactly as predicted for a utility-maximizing decision-maker with belief p . Both experiments thus establish that the self-reported posterior closely matches the model’s operative belief in decision-making.

We likewise validate the oracle’s ability to generate messages of a prescribed precision. For each oracle-generated message of specified precision q , an independent LLM produces an estimate $\hat{q}$ of the message’s precision. We repeat this for $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ over 100 programmatically generated profiles with 10 trials per combination, totaling 6,000 trials. The mean absolute error $|q - \hat{q}|$ is 0.025, indicating that the realized message precision closely tracks the requested precision.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

We now examine belief trajectories from a representative instance of the LLM experiment to characterize the effect of non-Bayesian belief updating. We fix the policy to the analytically optimal policy and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0, whereas the LLM belief (solid black line) is markedly more persistent. This macro-level behavior is a direct consequence of the micro-level biases: the public belief resists entering a cascade (NB3) because individual agents overreact (NB2) to the occasional positive signal that runs counter to the group’s increasingly negative belief.

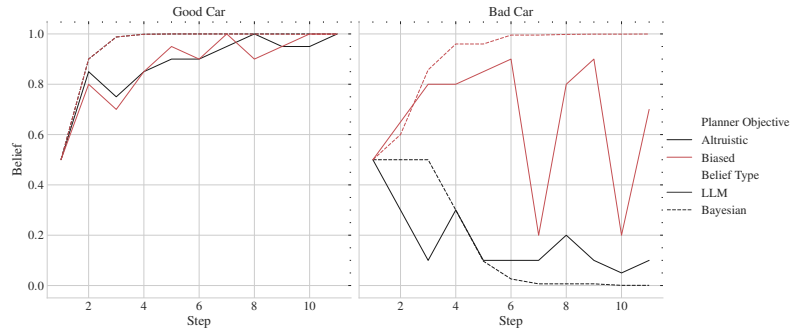


Figure 5: Belief-updating trajectories of the true Bayesian belief and the LLM-reported belief under the analytically optimal policy, across planner objectives and true states of the world.